SIMULTANEOUS CONFIDENCE BANDS FOR FUNCTIONAL VARYING-COEFFICIENT MODELS

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Supplementary Material

This supplement uses the notation of the main article. Sections S1–S2 prove the main results under the assumptions stated in the main article. Section S3 discusses sampling frequency, dimension, bandwidth, and implementation. Section S4 extends the analysis to high-dimensional functional means. Section S5 reports computational details and additional results.

S1. Linearization and covariance bounds

We use Assumptions 1–5 from the main article. Constants are uniform in n, p, h and the observation counts; r and q are fixed, and $s = r + 1$. Write $P = (r + 1)p$ and use $\|\cdot\|_{\max}$ for the largest absolute entry of a vector

or matrix at one time point. Estimation, residuals, and inference all use $\mathcal{T} = [0, 1]$. All other notation is as in Sections 2 and 3.

We use the scales in (3.1), (3.2), and (3.5), and the local basis and block selector from Section 2. Set

$$\mathbf{S}_i(t) = m_i^{-1} \sum_j K_h(T_{ij} - t)\boldsymbol{\psi}_{ij}(t)\boldsymbol{\psi}_{ij}(t)^\intercal, \qquad \widehat{\mathbf{D}}_h = n^{-1} \sum_i \mathbf{S}_i,$$

$$\mathbf{u}_i(t) = m_i^{-1} \sum_j K_h(T_{ij} - t)\boldsymbol{\psi}_{ij}(t)\varepsilon_i(T_{ij}), \qquad \bar{\mathbf{u}} = n^{-1} \sum_i \mathbf{u}_i,$$

$$\mathbf{D}_h(t) = \mathbb{E}\{\widehat{\mathbf{D}}_h(t)\}, \qquad \boldsymbol{\Psi}_i(t) = \mathbf{L}\mathbf{D}_h(t)^{-1}\mathbf{u}_i(t).$$

S1.1 Population design and uniform concentration

Lemma S1.1 (Population design)**.** *Under Assumptions 1, 2, and 3, uniformly on $\mathcal{T}$,*

$$c_D\mathbf{I}_P \preceq \mathbf{D}_h(t) \preceq C_D\mathbf{I}_P, \qquad \|\mathbf{D}_h(t) - \mathbf{D}_h(u)\|_{\mathrm{op}} \le Ch^{-1}|t - u|.$$

Define the kernel-window moment matrix by

$$I_{t,h} = [-1, 1] \cap [-t/h, (1 - t)/h],$$

$$\boldsymbol{\Gamma}_r(t, h) = \int_{I_{t,h}} \boldsymbol{\theta}_r(z)\boldsymbol{\theta}_r(z)^\intercal K(z)\, dz.$$

Then, uniformly for $t \in \mathcal{T}$,

$$\mathbf{D}_h(t) = \boldsymbol{\Gamma}_r(t, h) \otimes \{f(t)\boldsymbol{\Omega}(t)\} + O_{\mathrm{op}}(h).$$

Proof. The exact formula is

$$\mathbf{D}_h(t) = \int_{I_{t,h}} \boldsymbol{\theta}_r(z)\boldsymbol{\theta}_r(z)^\intercal K(z) \otimes \{f(t + hz)\boldsymbol{\Omega}(t + hz)\}\, dz.$$

For $h \le 1/2$, this interval contains either $[0,1]$ or $[-1,0]$. Continuity, symmetry, nonnegativity, and unit integral imply that K is positive on a nonempty interval in $(0,1)$ and on its reflection. The polynomial Gram matrix on either half is positive definite: a nonzero polynomial cannot vanish on such an interval. Its dimension $r+1$ is fixed. The eigenvalue bounds for $f\boldsymbol{\Omega}$ therefore give the asserted uniform lower bound; the upper bound follows from boundedness of the basis and the kernel. The L^1 increment, between evaluation points t and s, of each scalar kernel $u \mapsto K_h(u-t)((u-t)/h)^a$, $0 \le a \le 2r$, is at most $Ch^{-1}|t-s|$. Integrating against the operator-bounded $f\boldsymbol{\Omega}$ proves the increment bound. Operator-norm Lipschitz continuity of $f\boldsymbol{\Omega}$ on $\mathcal{T}$ gives the stated expansion with $\boldsymbol{\Gamma}_r(t,h)$, uniformly in t. The inverse matrices have the same uniform operator bound and an $O(h^{-1})$ Lipschitz constant by the resolvent identity. □

For the Taylor remainder, put

$$\mathbf{r}(t,u) = \boldsymbol{\beta}(u) - \sum_{a=0}^{r} \frac{\boldsymbol{\beta}^{(a)}(t)}{a!}(u-t)^a,$$

$$\mathbf{a}_i(t) = m_i^{-1} \sum_j K_h(T_{ij}-t)\boldsymbol{\psi}_{ij}(t)\mathbf{X}_{ij}^{\mathsf{T}}\mathbf{r}(t,T_{ij}),$$

and $\mathbf{a}_n = n^{-1}\sum_i \mathbf{a}_i$. Each coordinate of $\mathbf{r}$ is bounded by Ch^s on the kernel support, so $\|\mathbf{r}(t,u)\|_2 \le C\sqrt{p}\,h^s$ there.

Lemma S1.2 (Uniform first and second moments)**.** *Under Assumptions 1–4, define*

$$\mathcal{D}_n = \sup_t \|\widehat{\mathbf{D}}_h(t) - \mathbf{D}_h(t)\|_{\mathrm{op}}, \quad \mathcal{U}_n = \|\bar{\mathbf{u}}\|_{\infty,\mathcal{T}},$$

$$\mathcal{A}_n = \frac{\|\mathbf{a}_n - \mathbb{E}\{\mathbf{a}_n\}\|_{2,\mathcal{T}}}{\sqrt{p}\, h^s}.$$

Then

$$\mathbb{E}(\mathcal{D}_n^2 + \mathcal{A}_n^2) \le C\rho_n^2, \qquad \mathbb{E}(\mathcal{U}_n^2) \le C\delta_n^2.$$

Moreover,

$$\|\mathbb{E}\{\mathbf{a}_n\}\|_{2,\mathcal{T}} \le C\sqrt{p}\, h^s.$$

Proof. We first obtain a matrix variance bound from the existing coordinate moments and the upper eigenvalue bound for $\mathbf{\Omega}$. For every $\|\mathbf{v}\|_2 = 1$, Minkowski's inequality gives $\|\mathbf{v}^\top \mathbf{X}(u)\|_{L^{2q}} \le C\sqrt{p}$, whereas $\|\mathbf{v}^\top \mathbf{X}(u)\|_{L^2} \le C$. Interpolation and Hölder's inequality therefore imply

$$\mathbb{E}\{X_k(u)^2 (\mathbf{v}^\top \mathbf{X}(u))^2\} \le C\|\mathbf{v}^\top \mathbf{X}(u)\|^2_{L^{2q/(q-1)}} \le Cp^{1/(q-1)},$$

$$\left\|\mathbb{E}\{\|\mathbf{X}(u)\|_2^2 \mathbf{X}(u)\mathbf{X}(u)^\top\}\right\|_{\mathrm{op}} \le Cp^{q/(q-1)}.$$

These bounds hold uniformly in u. The bounded, fixed-dimensional local polynomial basis preserves them up to a constant.

Condition on a subject's trajectories and separate identical from distinct observation indices in the second moments. Identical indices contribute a kernel integral of order h^{-1}; distinct indices contribute bounded double integrals. For the latter, the squared conditional mean is bounded

by the weighted integral of the squared mark. Explicitly, for a nonnegative weight w and self-adjoint matrix $A(u)$,

$$\left\{\int w(u)A(u)\,du\right\}^2 \preceq \left\{\int w(u)\,du\right\}\int w(u)A(u)^2\,du.$$

Apply this with $w(u) = K_h(u-t)f(u)$ and the matrix mark $\boldsymbol{\theta}_r\boldsymbol{\theta}_r^{\mathsf{T}} \otimes \mathbf{X}_i(u)\mathbf{X}_i(u)^{\mathsf{T}}$; the vector version applies to the Taylor mark. The matrix moment bound and $\|\mathbf{r}(t,u)\|_2 \le C\sqrt{p}\,h^s$ thus give

$$\begin{aligned}\left\|\mathbb{E}\{(\mathbf{S}_i(t)-\mathbb{E}\mathbf{S}_i(t))^2\}\right\|_{\mathrm{op}} &\le Cp^{q/(q-1)}\{1+(m_ih)^{-1}\},\\ \frac{\mathbb{E}\|\mathbf{a}_i(t)-\mathbb{E}\mathbf{a}_i(t)\|_2^2}{ph^{2s}} &\le Cp^{q/(q-1)}\{1+(m_ih)^{-1}\}.\end{aligned}$$

The subject envelopes of $\mathbf{S}_i$ in spectral norm and of $\mathbf{a}_i/(\sqrt{p}\,h^s)$ in Euclidean norm are at most $Ch^{-1}\sup_u\|\mathbf{X}_i(u)\|_2^2$. Minkowski's inequality gives

$$\left\|\sup_u\|\mathbf{X}_i(u)\|_2^2\right\|_{L^q} \le \sum_{k=1}^{p}\left\|\sup_u|X_{ik}(u)|^2\right\|_{L^q} \le Cp.$$

Use a uniform grid χ_n of mesh at most

$$\Delta_n = n^{-8}h^{10}(pK_n)^{-8},$$

Then $\log(2P|\chi_n|) \le CL_n$. For each subject, form a block diagonal matrix whose blocks are the centered design matrices at all grid points. Its spectral norm is the largest block norm, and the matrix variance parameter of the sum is at most $Cnp^{q/(q-1)}v_n^2$. Its largest summand has L^2 norm at most $Cpn^{1/q}/h$. The second-moment inequality in Theorem I of Tropp (2016),

divided by n, yields

$$\left[\mathbb{E}\max_{t \text{ on grid}} \|\widehat{\mathbf{D}}_h(t) - \mathbf{D}_h(t)\|_{\text{op}}^2\right]^{1/2}$$
$$\leq C\left\{p^{q/(2q-2)} v_n \sqrt{L_n/n} + \frac{pL_n}{n^{1-1/q}h}\right\} = C\rho_n.$$

Apply the rectangular version of the same inequality to a block diagonal matrix with column blocks $\{\mathbf{a}_i(t) - \mathbb{E}\mathbf{a}_i(t)\}/(\sqrt{p}\, h^s)$. Both matrix variance parameters are bounded by the displayed vector second moments. This proves the same grid bound for $\mathcal{A}_n$.

For the score, every coordinate has envelope CF_i/h and second moment at most $C\{1 + (m_i h)^{-1}\}$. Applying Lemmas E.1 and E.2(ii) of Chernozhukov, Chetverikov and Kato (2017) simultaneously to the score coordinates and grid points gives expected maximum at most $C\delta_n$ and, above twice that expectation, the tail bound

$$\exp(-cnx^2/v_n^2) + \frac{Cn^{1-q}K_n^q}{h^q x^q}, \qquad x > 0.$$

Integrating, with the polynomial term capped at one, gives the score second moment bound because $q > 2$.

The samplewise Lipschitz constants of the design in spectral norm and of $\mathbf{a}_i/(\sqrt{p}\, h^s)$ in Euclidean norm are at most $CpF_i h^{-2}$; score coordinates have constants $CF_i h^{-2}$. For the Taylor term, use $|\partial_t r_k(t,u)| \leq C|u-t|^r$ almost everywhere on the kernel support; for $r = 0$, use the Lipschitz property of β_k. The compactly supported Lipschitz kernel gives the same

increment bound across the support boundary. Thus interpolation errors have L^2 norm at most $CpK_n\Delta_n h^{-2}$, which is absorbed by the respective rates. Finally,

$$\mathbb{E}\{\mathbf{a}_n(t)\} = \int K_h(u-t)\{\boldsymbol{\theta}_r((u-t)/h) \otimes \boldsymbol{\Omega}(u)\}\mathbf{r}(t,u)f(u)\,du.$$

The operator bound for $\boldsymbol{\Omega}$, the bounded basis, and $\|\mathbf{r}(t,u)\|_2 \le C\sqrt{p}\,h^s$ prove the last assertion. □

Lemma S1.3 (Uniform linear representation)**.** *Suppose Assumptions 1–4 hold and $\rho_n \to 0$. With probability tending to one, $\widehat{\mathbf{D}}_h$ is invertible throughout $\mathcal{T}$, and*

$$\widehat{\boldsymbol{\beta}}(t) - \boldsymbol{\beta}(t) = n^{-1}\sum_i \boldsymbol{\Psi}_i(t) + \mathbf{R}_n(t), \qquad t \in \mathcal{T}.$$

Uniformly on this domain,

$$\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_2 = O_P\{\sqrt{p}(\delta_n + h^s)\}, \qquad \|\mathbf{R}_n\|_{\max} = O_P\{\sqrt{p}(\rho_n\delta_n + h^s)\}.$$

More precisely, there is an event $\mathcal{E}_n$ with $\mathbb{P}(\mathcal{E}_n^c) \le C\rho_n^2$ such that, on this event,

$$\sup_t \|\widehat{\mathbf{D}}_h(t)^{-1}\|_{\mathrm{op}} \le C, \qquad \|\widehat{\mathbf{D}}_h^{-1} - \mathbf{D}_h^{-1}\|_{\mathrm{op},\mathcal{T}} \le C\mathcal{D}_n,$$

$$\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_{2,\mathcal{T}} \le C\sqrt{p}(\mathcal{U}_n + h^s), \qquad \|\mathbf{R}_n\|_{\infty,\mathcal{T}} \le C\sqrt{p}\,\mathcal{D}_n\mathcal{U}_n + C\sqrt{p}\,h^s.$$

Proof. Let

$$\mathcal{E}_n = \{\mathcal{D}_n \le c_D/2, \quad \mathcal{A}_n \le 1\}.$$

Lemma S1.2 and Markov's inequality give its exceptional probability. Weyl's inequality and the resolvent identity give the two inverse bounds. On $\mathcal{E}_n$,

$$\|\mathbf{a}_n\|_{2,\mathcal{T}} \le C\sqrt{p}\,h^s + \sqrt{p}\,h^s\mathcal{A}_n \le C\sqrt{p}\,h^s.$$

The exact normal equation is

$$\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \mathbf{L}\widehat{\mathbf{D}}_h^{-1}(\bar{\mathbf{u}} + \mathbf{a}_n).$$

Here the Taylor coefficients are stacked with their factors $h^a/a!$, so their first block is exactly $\boldsymbol{\beta}(t)$. Since $\|\bar{\mathbf{u}}\|_2 \le \sqrt{P}\,\mathcal{U}_n$, the estimation bound follows. Subtracting $\mathbf{L}\mathbf{D}_h^{-1}\bar{\mathbf{u}}$ yields

$$\mathbf{R}_n = \mathbf{L}(\widehat{\mathbf{D}}_h^{-1} - \mathbf{D}_h^{-1})\bar{\mathbf{u}} + \mathbf{L}\widehat{\mathbf{D}}_h^{-1}\mathbf{a}_n.$$

Its first term is at most $C\sqrt{p}\,\mathcal{D}_n\mathcal{U}_n$ in Euclidean norm, and hence in maximum norm. The second is at most $C\sqrt{p}\,h^s$ on $\mathcal{E}_n$. Lemma S1.2 also gives $\mathbb{E}(\mathcal{D}_n\mathcal{U}_n) \le C\rho_n\delta_n$ by Cauchy–Schwarz. This proves all assertions and the quantitative transfer bound used below. □

The preceding lemma proves Theorem 1.

S1.2 Covariance and its sampling components

Lemma S1.4 (Covariance identity and variance order)**.** *Under Assumptions 1–5, define*

$$\mathbf{A}_h(t,u) = \mathbf{L}\mathbf{D}_h(t)^{-1}\{\boldsymbol{\theta}_r((u-t)/h) \otimes \mathbf{I}_p\}K_h(u-t).$$

Then the covariance of $n^{-1/2}\sum_i \boldsymbol{\Psi}_i$ *is*

$$\begin{aligned}
\mathbf{V}_n(s,t) &= \kappa_{1,n}\mathbf{V}_{h,d}(s,t) + (1-\kappa_{1,n})\mathbf{V}_{h,o}(s,t),\\
\mathbf{V}_{h,d}(s,t) &= \int \mathbf{A}_h(s,u)\mathbf{C}(u,u)\mathbf{A}_h(t,u)^\top f(u)\,du,\\
\mathbf{V}_{h,o}(s,t) &= \iint \mathbf{A}_h(s,u)\mathbf{C}(u,v)\mathbf{A}_h(t,v)^\top f(u)f(v)\,du\,dv.
\end{aligned}$$

Let $\mathbf{e}_0 = (1,0,\ldots,0)^\top$ *and define*

$$\begin{aligned}
K_{r,t,h}(z) &= \mathbf{e}_0^\top \boldsymbol{\Gamma}_r(t,h)^{-1}\boldsymbol{\theta}_r(z)K(z)\mathbf{1}_{I_{t,h}}(z),\\
\nu_r(t,h) &= \int K_{r,t,h}(z)^2\,dz.
\end{aligned}$$

There are constants $0 < c_\nu < C_\nu$ *such that* $c_\nu \le \nu_r(t,h) \le C_\nu$ *uniformly in* t, h*. Uniformly on* $\mathcal{T}$*,*

$$\mathbf{V}_n(t,t) = \left\{1 - \kappa_{1,n} + \frac{\kappa_{1,n}\nu_r(t,h)}{hf(t)}\right\}\mathbf{V}_F(t) + O_{\mathrm{op}}(hv_n^2).$$

In particular, with $\tau_k^2(t) = [\mathbf{V}_n(t,t)]_{kk}$*,*

$$cv_n^2 \le \tau_k^2(t) \le Cv_n^2 \qquad (t \in \mathcal{T}, k \le p).$$

Proof. Expand the two subject sums. Identical observation indices have score second moment $\mathbf{C}(u,u)$; distinct indices have joint moment $\mathbf{C}(u,v)$

and independent times with density $f(u)f(v)$. There are m_i pairs of the first type and $m_i(m_i-1)$ of the second. Averaging the identity over subjects gives the result. Both covariance components arise from the single functional error process; their distinction concerns observation indices.

Lemma S1.1 and the operator inverse identity give, uniformly for $t \in \mathcal{T}$ and $z \in I_{t,h}$,

$$h\mathbf{A}_h(t, t+hz) = \frac{K_{r,t,h}(z)}{f(t)}\mathbf{\Omega}(t)^{-1} + O_{\mathrm{op}}(h).$$

The moment identity

$$\mathbf{\Gamma}_r(t,h)\mathbf{e}_0 = \int_{I_{t,h}} \boldsymbol{\theta}_r(z)K(z)\,dz$$

implies $\int K_{r,t,h} = 1$. The uniform inverse bound for $\mathbf{\Gamma}_r(t,h)$ gives $\nu_r(t,h) \leq C$; Cauchy–Schwarz and $|I_{t,h}| \leq 2$ give $\nu_r(t,h) \geq 1/2$. Changing variables in the two covariance integrals and using operator-norm Lipschitz continuity gives, uniformly on $\mathcal{T}$,

$$\mathbf{V}_{h,d}(t,t) = \frac{\nu_r(t,h)}{hf(t)}\mathbf{V}_F(t) + O_{\mathrm{op}}(1),$$

$$\mathbf{V}_{h,o}(t,t) = \mathbf{V}_F(t) + O_{\mathrm{op}}(h).$$

The weighted error is at most $C(h+\kappa_{1,n}) = Chv_n^2$. The coordinate variances of $\mathbf{V}_F$ are bounded above by the operator bounds and below by assumption. The scalar coefficient in braces is comparable to $1 + \kappa_{1,n}/h$, uniformly over $0 \leq \kappa_{1,n} \leq 1$. Absorbing the $O(hv_n^2)$ error proves the variance bounds. $\square$

Write

$$\mathbf{T}(t) = \mathrm{diag}(\tau_1^2(t), \ldots, \tau_p^2(t)), \qquad \mathbf{M}_i(t) = \mathbf{T}(t)^{-1/2}\mathbf{\Psi}_i(t),$$

and $\mathbf{Q}_n(s,t) = n^{-1}\sum_i \mathbb{E}\{\mathbf{M}_i(s)\mathbf{M}_i(t)^\intercal\}$. Thus each coordinate of the matching Gaussian process has variance one, without any requirement that its full covariance be invertible.

S1.3 Oracle moments and empirical covariance

Lemma S1.5 (Subject moments and oracle covariance)**.** *Under Assumptions 1–5,*

$$\max_i \left\| \sup_{t\in\mathcal{T}} \|\mathbf{M}_i(t)\|_\infty \right\|_{L^q} \le \frac{C\sqrt{p}K_n}{hv_n}, \qquad \sup_{t,k} \frac{1}{n}\sum_i \mathbb{E}\{|M_{ik}(t)|^4\} \le Cp^2\Lambda_n.$$

Moreover,

$$\mathbb{E}\{R_{Q,n}\} \le Ce_n, \qquad R_{Q,n} = \left\| n^{-1}\sum_i \mathbf{M}_i(s)\mathbf{M}_i(t)^\intercal - \mathbf{Q}_n(s,t) \right\|_{\infty,\mathcal{T}^2}.$$

For

$$J_{i,kl}(t) = m_i^{-1}\sum_j K_h(T_{ij}-t)|X_{ijk}X_{ijl}|,$$

define

$$\mathcal{J}_n = \max\left\{ \sup_{t,a} \frac{1}{nv_n^2}\sum_i u_{ia}(t)^2, \sup_{t,k,l} \frac{1}{nv_n^2}\sum_i J_{i,kl}(t)^2 \right\}.$$

There is a fixed C_J for which $\mathbb{P}(\mathcal{J}_n > C_J) \le Ce_n$.

Proof. Every row of $\mathbf{L}\mathbf{D}_h^{-1}$ has uniformly bounded Euclidean norm, so its sum of absolute entries is at most $C\sqrt{p}$. Since every score coordinate is bounded by CF_i/h, this proves the envelope bound. For any scalar score entry, expanding its fourth moment into groups with $d \in \{1, 2, 3, 4\}$ distinct observation indices gives

$$\mathbb{E}\{|u_{ia}(t)|^4\} \le C\{1 + (m_i h)^{-3}\}.$$

Indeed, a time appearing b times contributes $\int |K_h(u-t)|^b f(u)\, du \le Ch^{1-b}$, mixed fourth mark moments are uniformly bounded by Hölder's inequality, and the group contains at most Cm_i^d terms. Its contribution is at most $C(m_i h)^{d-4}$. The same argument applies to $J_{i,kl}$. Minkowski's inequality and the $C\sqrt{p}$ row-sum bound therefore give a fourth moment at most $Cp^2\{1 + (m_i h)^{-3}\}$ for each Ψ_{ik}. Divide by $\tau_k^4 \asymp v_n^4$ and average.

For covariance products the average coordinate variance is at most $Cp^2\Lambda_n$, by Cauchy–Schwarz and the last bound. Their common envelope is $CpF_i^2/(h^2 v_n^2)$, whose maximum over subjects has L^2 norm at most

$$\frac{Cp}{h^2 v_n^2}\left(\sum_i \mathbb{E}\{F_i^q\}\right)^{2/q} \le \frac{Cpn^{2/q}K_n^2}{h^2 v_n^2}.$$

Applying the same expectation maximal inequality to all signed coordinate products and pairs of grid points gives

$$\mathbb{E}R_{Q,n} \le C\left\{p\sqrt{\Lambda_n L_n/n} + \frac{pK_n^2 L_n}{n^{1-2/q}h^2 v_n^2}\right\} \le Ce_n.$$

For completeness, interpolation here can also be controlled uniformly in p. The weights satisfy

$$\int \|\mathbf{A}_h(t,u) - \mathbf{A}_h(s,u)\|_{\mathrm{op}}^2 f(u)\,du \le Ch^{-3}|t-s|^2,$$
$$\int \|\mathbf{A}_h(t,u) - \mathbf{A}_h(s,u)\|_{\mathrm{op}} f(u)\,du \le Ch^{-1}|t-s|.$$

These follow from the kernel increments and the $O(h^{-1})$ Lipschitz bound for $\mathbf{D}_h^{-1}$. The covariance identity and the operator bound for $\mathbf{C}$ imply

$$\frac{1}{n}\sum_i \mathbb{E}\{|\Psi_{ik}(t) - \Psi_{ik}(s)|^2\} \le Cv_n^2 h^{-2}|t-s|^2.$$

The reverse triangle inequality in this L^2 norm gives $|\tau_k(t) - \tau_k(s)| \le Cv_n h^{-1}|t-s|$. Consequently, samplewise on $\mathcal{T}$,

$$\sup_k |M_{ik}(t) - M_{ik}(s)| \le \frac{C\sqrt{p}F_i}{v_n h^2}|t-s|.$$

The expected covariance-product interpolation error is therefore at most $CpK_n^2\Delta_n/(h^3 v_n^2)$, which is absorbed by e_n, since $e_n \ge p^{3/2}\delta_n \ge \sqrt{L_n/n}$.

Finally, each coordinate of $\mathbf{u}_i/v_n$ and $J_{i,kl}/v_n$ has average second moment bounded by a constant. Their fourth moments and product envelopes obey the same calculation with the row-sum factor omitted (while keeping L_n and K_n). Thus the expected maximum fluctuation of their empirical squares is at most

$$C\left\{\sqrt{\Lambda_n L_n/n} + \frac{K_n^2 L_n}{n^{1-2/q}h^2 v_n^2}\right\} \le Ce_n.$$

Markov's inequality at a fixed threshold above the population second-moment bound proves the assertion about $\mathcal{J}_n$. □

S1.4 Fitted residuals and covariance control

Let $\widehat{\varepsilon}_{ij} = Y_{ij} - \mathbf{X}_{ij}^{\top}\widehat{\boldsymbol{\beta}}(T_{ij})$, and define $\widehat{\mathbf{u}}_i$ by replacing $\varepsilon_i(T_{ij})$ in $\mathbf{u}_i$ by this fitted residual. Put $\widehat{\boldsymbol{\Psi}}_i = \mathbf{L}\widehat{\mathbf{D}}_h^{-1}\widehat{\mathbf{u}}_i$ and

$$\widehat{\sigma}_k^2(t) = n^{-1}\sum_i \widehat{\Psi}_{ik}(t)^2, \quad \widehat{\mathbf{Q}}_n(s,t)_{kl} = \frac{n^{-1}\sum_i \widehat{\Psi}_{ik}(s)\widehat{\Psi}_{il}(t)}{\widehat{\sigma}_k(s)\widehat{\sigma}_l(t)}.$$

Let $\mathcal{V}_n$ be the event that $\widehat{\mathbf{D}}_h$ is invertible on $\mathcal{T}$ and all fitted coordinate variances are positive on $\mathcal{T}$. On this event define

$$\vartheta_n = \sup_{t,k}\left|\frac{\widehat{\sigma}_k(t)}{\tau_k(t)} - 1\right|, \qquad \Delta_{Q,n} = \|\widehat{\mathbf{Q}}_n - \mathbf{Q}_n\|_{\infty,\mathcal{T}^2}.$$

Set both quantities to $+\infty$ on $\mathcal{V}_n^c$.

Lemma S1.6 (Quantitative feasible covariance)**.** *Under Assumptions 1–5, if $e_n \to 0$, then, for all sufficiently large n and $0 < u \le 1/2$,*

$$\mathbb{P}\{\vartheta_n + \Delta_{Q,n} > u\} \le Ce_n/u.$$

In particular, $\mathbb{P}(\mathcal{V}_n^c) \le Ce_n$ and the discrepancies are $O_P(e_n)$. The fitted-score replacement obeys

$$\sup_{t,k}\left[\frac{1}{n}\sum_i\left\{\frac{\widehat{\Psi}_{ik}(t) - \Psi_{ik}(t)}{\tau_k(t)}\right\}^2\right]^{1/2} = O_P\{p^{3/2}(\delta_n + h^s)\}.$$

Proof. The condition $e_n \to 0$ implies $p\delta_n \to 0$ and hence $\rho_n \to 0$, so Lemma S1.3 applies. Define

$$\mathcal{H}_n = \mathcal{E}_n \cap \{R_{Q,n} \le 1,\ \mathcal{J}_n \le C_J\}.$$

Its complement has probability at most $C(\rho_n^2 + e_n)$. Let $b_n^{\mathrm{fit}} = \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_{2,\mathcal{T}}$ and $J_i^0(t) = \sum_k J_{i,kk}(t)$. Uniform empirical invertibility gives, on $\mathcal{H}_n$,

$$\sup_{t,k} \sum_{a=1}^{P} |[\mathbf{L}\widehat{\mathbf{D}}_h(t)^{-1}]_{ka}| \le \sqrt{P} \sup_t \|\mathbf{L}\widehat{\mathbf{D}}_h(t)^{-1}\|_{\mathrm{op}} \le C\sqrt{p}.$$

The residual identity and weighted Cauchy–Schwarz give, for a score coordinate a corresponding to predictor coordinate k,

$$|\widehat{u}_{ia}(t) - u_{ia}(t)| \le b_n^{\mathrm{fit}} \{J_{i,kk}(t) J_i^0(t)\}^{1/2}.$$

Since $J_i^0 = \sum_l J_{i,ll}$, the definition of $\mathcal{J}_n$ implies

$$\sup_t \frac{1}{n v_n^2} \sum_i \|\mathbf{u}_i(t)\|_2^2 \le Cp,$$

$$\sup_{t,a} \frac{1}{n v_n^2} \sum_i |\widehat{u}_{ia}(t) - u_{ia}(t)|^2 \le Cp (b_n^{\mathrm{fit}})^2.$$

Indeed, each summand satisfies

$$n^{-1} \sum_i J_{i,kk} J_{i,ll} \le \{n^{-1} \sum_i J_{i,kk}^2\}^{1/2} \{n^{-1} \sum_i J_{i,ll}^2\}^{1/2} \le C v_n^2.$$

Use the exact decomposition

$$\widehat{\boldsymbol{\Psi}}_i - \boldsymbol{\Psi}_i = \mathbf{L}(\widehat{\mathbf{D}}_h^{-1} - \mathbf{D}_h^{-1})\mathbf{u}_i + \mathbf{L}\widehat{\mathbf{D}}_h^{-1}(\widehat{\mathbf{u}}_i - \mathbf{u}_i).$$

Applying the empirical L^2 norm to each coordinate and using the empirical row-sum bound gives a fitted-score discrepancy at most

$$\zeta_n = C\sqrt{p}\,\mathcal{D}_n + Cp^{3/2}(\mathcal{U}_n + h^s),$$

where we used $b_n^{\mathrm{fit}} \le C\sqrt{p}(\mathcal{U}_n + h^s)$. Lemma S1.2 and $\rho_n \lesssim p\delta_n$ give

$$\mathbb{E}\zeta_n \le Cp^{3/2}(\delta_n + h^s), \qquad \mathbb{E}\zeta_n^2 \le Cp^3(\delta_n + h^s)^2.$$

No independence of fitted residuals and the fitting observations is used in these inequalities.

Before dividing by fitted variances, write

$$\mathcal{W}_n = \sup_{s,t,k,l} \left| \frac{n^{-1}\sum_i \widehat{\Psi}_{ik}(s)\widehat{\Psi}_{il}(t)}{\tau_k(s)\tau_l(t)} - [\mathbf{Q}_n(s,t)]_{kl} \right|.$$

On $\mathcal{H}_n$, the empirical oracle coordinate second moments are at most 2, because $R_{Q,n} \le 1$. Expanding the difference of products and applying Cauchy–Schwarz therefore gives

$$\begin{aligned} \mathcal{W}_n &\le R_{Q,n} + C\zeta_n + \zeta_n^2, \\ \mathbb{E}\{\mathbf{1}_{\mathcal{H}_n}\mathcal{W}_n\} &\le C\{e_n + p^3(\delta_n + h^s)^2\} \\ &\le Ce_n. \end{aligned}$$

The integrand is defined to be zero outside $\mathcal{H}_n$. If $\mathcal{W}_n$ is smaller than a fixed constant, its diagonal entries give positive fitted variances and fitted-to-oracle variance ratios in $[1/2, 3/2]$. The square-root and inverse square-root

maps are Lipschitz on this interval. Since $|[\mathbf{Q}_n(s,t)]_{kl}| \le 1$,

$$\vartheta_n + \Delta_{Q,n} \le C\mathcal{W}_n$$

on this event. Consequently, for $0 < u \le 1/2$, Markov's inequality and the exceptional-probability bound give

$$\mathbb{P}\{\vartheta_n + \Delta_{Q,n} > u\} \le C(\rho_n^2 + e_n) + Ce_n/u \le Ce_n/u.$$

The last step uses $\rho_n^2 \lesssim p^2\delta_n^2 \le p^3(\delta_n + h^s)^2 \le e_n$ for all sufficiently large n. This proves the tail bound, and the fitted-score rate follows from its samplewise bound on $\mathcal{H}_n$ and the vanishing exceptional probability. □

S2. Gaussian and bootstrap approximation

Recall $L_n = \log(np/h)$, $s = r + 1$, and let $\mathcal{F}_n$ denote the sigma-field generated by the observations. All conditional probabilities and expectations below condition on this sigma-field. Put

$$M_{ik}(t) = \Psi_{ik}(t)/\tau_k(t), \qquad Z_{n,k}(t) = n^{-1/2}\sum_i M_{ik}(t),$$

and write

$$Q_n(u,t) = n^{-1}\sum_i \mathbb{E}\{M_i(u)M_i(t)^\intercal\}.$$

The matching Gaussian process G_n, its supremum $S_{G,n}$, the statistics T_n°, T_n, and the multiplier process and maximum G_n^*, S_n^* are those defined in Section 3.

The preceding lemmas give the following inputs. With $F_i = 1 + \sup_t \|\mathbf{X}_i(t)\|_\infty^2 + \sup_t \|\mathbf{X}_i(t)\varepsilon_i(t)\|_\infty$, we have $\|F_i\|_q \le CK_n$, where $1 \le K_n \le Cp^{1/q}$. Lemma S1.4 gives $\tau_k(t) \asymp v_n$ uniformly. The subject moment bounds imply

$$\left\| \max_k \sup_t |M_{ik}(t)| \right\|_q \le CH_n, \qquad \max_k \sup_t \frac{1}{n} \sum_i \mathbb{E}\{M_{ik}(t)^4\} \le CA_n, \tag{S2.1}$$

where

$$H_n = \frac{\sqrt{p} K_n}{h v_n}, \qquad A_n = p^2 \Lambda_n, \qquad B_n = \max\{H_n, \sqrt{A_n}\}.$$

The $\sqrt{p}$ row-sum bound for $\mathbf{L}\mathbf{D}_h^{-1}$ gives the score envelope, and Minkowski's inequality gives the fourth-moment factor p^2. These estimates use the general dimension bounds under the stated assumptions. Since $h^2 v_n^2 \le h^2 + h \le 3/4$, we have $H_n \ge 1$, so a separate constant in the maximum defining B_n is unnecessary.

Use γ_n, e_n from (3.2) and (3.5), and write

$$r_{n,1} = \frac{\sqrt{np}\, \rho_n \delta_n}{v_n}, \qquad r_{n,2} = \frac{\sqrt{np}\, h^s}{v_n}.$$

The quantitative version of Lemma S1.3, including the event controlling the centered Taylor score, gives, for every $u > 0$,

$$\mathbb{P}\left\{ \left\| \sqrt{n}\ \mathrm{diag}(\tau_k^{-1})(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - Z_n \right\|_{\infty,\mathcal{T}} > Cr_{n,2} + u \right\} \le C\rho_n^2 + Cr_{n,1}/u. \tag{S2.2}$$

In particular, the smoothing term in this bound is deterministic on an event whose complement has probability at most $C\rho_n^2$; an assertion in probability about the smoothing remainder alone would not suffice for the coverage calculation below.

Let $\mathcal{V}_n$ be the event of uniform empirical design invertibility and positive fitted variances. On its complement, extend T_n, S_n^*, G_n^* by zero and use empty bands. The statistic T_n° is extended by zero only when the empirical design fails to be uniformly invertible. On $\mathcal{V}_n$, write

$$\vartheta_n = \max_k \sup_t \left| \frac{\widehat{\sigma}_k(t)}{\tau_k(t)} - 1 \right|, \qquad \Delta_{Q,n} = \|\widehat{Q}_n - Q_n\|_{\infty,\mathcal{T}^2},$$

where $\widehat{Q}_n(s,t) = n^{-1}\sum_i \widehat{M}_i(s)\widehat{M}_i(t)^\intercal$ and $\widehat{M}_{ik} = \widehat{\Psi}_{ik}/\widehat{\sigma}_k$. Set both discrepancies to $+\infty$ on $\mathcal{V}_n^c$. Lemma S1.6 supplies the quantitative bound

$$\mathbb{P}(\vartheta_n + \Delta_{Q,n} > u) \le Ce_n/u, \qquad 0 < u \le 1/2. \tag{S2.3}$$

This is a bound on an unconditional probability over the observations; it is stronger than the corresponding $O_P(e_n)$ assertion.

S2.1 A fine grid and continuous Gaussian processes

Choose a regular grid $\chi_n \subset \mathcal{T}$ containing the endpoints and having mesh at most

$$\Delta_n = n^{-8}h^{10}(pK_n)^{-8}, \qquad \epsilon_n = n^{-2}L_n^{-1/2}.$$

Its cardinality $|\chi_n|$ can be chosen of order $1 + \Delta_n^{-1}$. Since $K_n \leq Cp^{1/q}$,

$$\log(2p|\chi_n|) \leq CL_n. \tag{S2.4}$$

Thus no polynomial lower bound on h is needed. Let $\pi_n(t)$ be the nearest grid point to the left, with the right endpoint mapped to itself, and put $\omega_n(W) = \max_k \sup_t |W_k(t) - W_k(\pi_n(t))|$.

Lemma S2.1 (Oracle increments and discretization)**.** *There is a continuous version of G_n. Moreover,*

$$\mathbb{E}\{\omega_n(Z_n)\} \leq C\sqrt{np}\, K_n h^{-2} \Delta_n / v_n, \tag{S2.5}$$

$$\mathbb{E}\{\omega_n(G_n)\} \leq Ch^{-1}\Delta_n \sqrt{L_n}, \tag{S2.6}$$

$$\mathbb{P}\{\omega_n(Z_n) > \epsilon_n\} + \mathbb{P}\{\omega_n(G_n) > \epsilon_n\} \leq Cn^{-1}. \tag{S2.7}$$

Proof. The kernel and population inverse satisfy the uniform bounds $\|\mathbf{D}_h(t)^{-1}\|_{\mathrm{op}} \leq C$ and $\|\mathbf{D}_h(t)^{-1} - \mathbf{D}_h(s)^{-1}\|_{\mathrm{op}} \leq C|t-s|/h$. These follow from population invertibility and the L^1 increment bound for the kernel weights, using the operator bounds on the design covariance. The score envelopes consequently give

$$\max_k |\Psi_{ik}(t)| \leq C\sqrt{p} F_i / h, \qquad \max_k |\Psi_{ik}(t) - \Psi_{ik}(s)| \leq C\sqrt{p} F_i h^{-2} |t-s|.$$

For completeness, the exact covariance calculation also gives

$$\frac{1}{n}\sum_i \mathbb{E}\{|\Psi_{ik}(t)-\Psi_{ik}(s)|^2\} \le C\{\kappa_{1,n}h^{-3}+h^{-2}\}|t-s|^2 = Cv_n^2h^{-2}|t-s|^2. \tag{S2.8}$$

To verify this, separate equal and distinct observation indices. For $|t-s| \le h$, the squared L^2 norm of the difference of the two coefficient-weight kernels is at most $Ch^{-3}|t-s|^2$, and its L^1 norm is at most $Ch^{-1}|t-s|$. The diagonal and off-diagonal covariance integrals are therefore bounded by the two terms in (S2.8), respectively, using the operator bound on the score covariance. For $|t-s| > h$, the variance bound $\tau_k^2 \le Cv_n^2$ yields the same inequality.

The reverse triangle inequality in the Hilbert space with squared norm $n^{-1}\sum_i \mathbb{E}\{(\cdot)^2\}$ now implies $|\tau_k(t)-\tau_k(s)| \le Cv_nh^{-1}|t-s|$. Together with $\tau_k \ge cv_n$, this gives

$$\max_k |M_{ik}(t)-M_{ik}(s)| \le C\sqrt{p}F_i(v_nh^2)^{-1}|t-s|, \tag{S2.9}$$

$$\frac{1}{n}\sum_i \mathbb{E}\{|M_{ik}(t)-M_{ik}(s)|^2\} \le Ch^{-2}|t-s|^2. \tag{S2.10}$$

Summing the first inequality and taking expectations proves (S2.5). The covariance Q_n is positive semidefinite because it is an average of covariance kernels. Thus it defines a Gaussian process. Equation (S2.10) and the Gaussian fourth-moment identity give a continuous version by the Kolmogorov continuity criterion.

Refine every interval of χ_n dyadically. At refinement level j, there are at most $Cp|\chi_n|2^j$ relevant coordinate increments, each with standard deviation at most $Ch^{-1}\Delta_n 2^{-j}$. The Gaussian maximal inequality and summation over $j \geq 0$ give

$$\mathbb{E}\{\omega_n(G_n)\} \leq Ch^{-1}\Delta_n \sum_{j\geq 0} 2^{-j}\sqrt{\log(2p|\chi_n|) + j} \leq Ch^{-1}\Delta_n\sqrt{L_n}.$$

Markov's inequality proves (S2.7). For example, $L_n \leq Cnp/h$, $p, K_n, v_n \geq 1$, and $h \leq 1$ show that each expectation in (S2.5)–(S2.6), divided by ϵ_n, is at most Cn^{-1}. □

The same grid suffices for the expected empirical covariance bounds. Indeed, (S2.9) and the subject envelopes bound increments of $M_{ik}(s)M_{il}(t)$ by $CpF_i^2(v_n^2h^3)^{-1}$ times the distance between the two time pairs. Their expected interpolation error is at most $CpK_n^2(v_n^2h^3)^{-1}\Delta_n$ and is absorbed by e_n. The logarithm of the number of coordinate-grid pairs is still $O(L_n)$.

Lemma S2.2 (Gaussian supremum bounds)**.** *The distribution function* $F_G(z) = \mathbb{P}(S_{G,n} \leq z)$ *is continuous and*

$$\mathbb{E}\{S_{G,n}\} \leq C\sqrt{L_n}, \qquad \sup_z \mathbb{P}(z < S_{G,n} \leq z + u) \leq Cu\sqrt{L_n} \quad (u > 0).$$

For $0 < u \leq 1/2$,

$$\sup_{z\geq 0} |F_G((1 \pm u)z) - F_G(z)| \leq C(uL_n + n^{-1}). \tag{S2.11}$$

Proof. Every coordinate variance equals one. The Gaussian maximal inequality on the signed grid and Lemma S2.1 give

$$\mathbb{E}\{S_{G,n}\} \leq \sqrt{2\log(2p|\chi_n|)} + \mathbb{E}\{\omega_n(G_n)\} \leq C\sqrt{L_n}.$$

Apply the equal-variance anti-concentration inequality in Theorem 3(i) of Chernozhukov, Chetverikov and Kato (2015) to signed coordinates on successively finer nested grids. Their maxima increase almost surely to $S_{G,n}$ and their expectations are bounded by $\mathbb{E}\{S_{G,n}\}$. Taking limits gives the displayed anti-concentration bound and hence continuity of F_G. The Gaussian union bound on χ_n, followed by (S2.7), gives $\mathbb{P}(S_{G,n} > C_0\sqrt{L_n}) \leq Cn^{-1}$ for a fixed sufficiently large C_0. For $z \leq 2C_0\sqrt{L_n}$, anti-concentration bounds a multiplicative shift by CuL_n. For larger z, both thresholds exceed $C_0\sqrt{L_n}$, giving (S2.11). □

S2.2 Proof of Theorem 2

Stack both signs of the values of M_i on χ_n. These independent, centered vectors have dimension $2p|\chi_n|$, average coordinate variances one, average coordinate fourth moments at most $CA_n \leq CB_n^2$, and qth moments of their maximum norm at most $CH_n^q \leq CB_n^q$. Theorem 2.5 of Chernozhukov et al.

(2022), applied to this signed vector, and (S2.4) therefore give

$$\sup_z |\mathbb{P}(\|Z_n\|_{\chi_n} \le z) - \mathbb{P}(\|G_n\|_{\chi_n} \le z)| \le C\left\{\left(\frac{B_n^2 L_n^5}{n}\right)^{1/4} + \frac{B_n L_n^{3/2-1/q}}{n^{1/2-1/q}}\right\} \le C\gamma_n.$$

This theorem permits singular full covariance matrices. Its variance condition is a coordinatewise condition, satisfied here by normalization. Lemma S2.1 replaces grid maxima by continuous suprema. The resulting shifts are at most $2\epsilon_n$, costing $C\epsilon_n\sqrt{L_n} = Cn^{-2}$ by Lemma S2.2. Since $B_n \ge 1$ and $L_n \ge \log 2$, the first term of γ_n absorbs n^{-1}. Consequently,

$$\sup_z |\mathbb{P}(\|Z_n\|_{\infty,\mathcal{T}} \le z) - F_G(z)| \le C\gamma_n. \tag{S2.12}$$

For the estimator, combine (S2.2) with (S2.12) and Gaussian anti-concentration. For any $u > 0$, the resulting bound is

$$C\{\gamma_n + \rho_n^2 + r_{n,1}/u + (r_{n,2} + u)\sqrt{L_n}\}.$$

Choose $u = (r_{n,1}/\sqrt{L_n})^{1/2}$. The definition of γ_n includes the resulting term $(r_{n,1}\sqrt{L_n})^{1/2}$ and the bias term $r_{n,2}\sqrt{L_n}$. Also,

$$r_{n,1}\sqrt{L_n} = \frac{\sqrt{npL_n}\,\rho_n\delta_n}{v_n} \ge \sqrt{p}\,L_n\rho_n,$$

because $\delta_n \ge v_n\sqrt{L_n/n}$. Since $\rho_n \to 0$, $\rho_n^2 \lesssim (r_{n,1}\sqrt{L_n})^{1/2}$ for all sufficiently large n. Thus the exceptional-event probability is absorbed, giving

$$\sup_z |\mathbb{P}(T_n^\circ \le z) - F_G(z)| \le C\gamma_n. \tag{S2.13}$$

S2.3 Conditional bootstrap discretization

Lemma S2.3 (Quantitative conditional discretization)**.** *Let*

$$r_n = \mathbb{P}\{\omega_n(G_n^*) > \epsilon_n \mid \mathcal{F}_n\},$$

with the zero extension already specified. Under the covariance and linearization rate conditions,

$$\mathbb{P}(r_n > n^{-1}) \le C(e_n + n^{-1}), \qquad r_n = O_P(n^{-1}). \tag{S2.14}$$

Proof. Let $b_n = \sup_{t\in\mathcal{T}} \|\widehat{\boldsymbol{\beta}}(t) - \boldsymbol{\beta}(t)\|_2$ and $\overline{F}_a = n^{-1}\sum_i F_i^a$ for $a = 1, 2$. Use the event $\mathcal{H}_n$ on which the empirical inverse is uniformly bounded, $\vartheta_n \le 1/2$, and $\sqrt{p}b_n \le 1$. The quantitative linearization and covariance lemmas give

$$\mathbb{P}(\mathcal{H}_n^c) \le Ce_n.$$

In particular, the truncated first moment bound $\mathbb{E}[\mathbf{1}_{\mathcal{E}_n} b_n] \le C\sqrt{p}(\delta_n + h^s)$ on the design event gives $\mathbb{P}(\mathcal{E}_n \cap \{\sqrt{p}b_n > 1\}) \le Cp(\delta_n + h^s) \le Ce_n$.

On $\mathcal{H}_n$, the samplewise residual identity gives

$$\|\mathbf{X}_{ij}\widehat{\varepsilon}_{ij}\|_2 \le \|\mathbf{X}_{ij}\varepsilon_{ij}\|_2 + \|\mathbf{X}_{ij}\|_2^2 b_n \le C\sqrt{p}F_i.$$

Residuals are fixed as the evaluation point varies. The kernel increment bound and the empirical inverse identity therefore imply

$$\|\widehat{\mathbf{u}}_i(t) - \widehat{\mathbf{u}}_i(s)\|_2 \le C\sqrt{p}F_i h^{-2}|t-s|,$$

$$\|\widehat{\mathbf{D}}_h(t)^{-1} - \widehat{\mathbf{D}}_h(s)^{-1}\|_{\mathrm{op}} \le Cp\overline{F}_1 h^{-2}|t-s|,$$

$$\max_k |\widehat{\Psi}_{ik}(t) - \widehat{\Psi}_{ik}(s)| \le Cp^{3/2}(1+\overline{F}_1)F_i h^{-3}|t-s|.$$

Thus, with $D_n = Cp^{3/2}(1+\overline{F}_1)\overline{F}_2^{1/2}h^{-3}$, the empirical L^2 norm of each fitted contribution increment is at most $D_n|t-s|$. The reverse triangle inequality for this empirical norm also gives $|\widehat{\sigma}_k(t) - \widehat{\sigma}_k(s)| \le D_n|t-s|$. Since $\widehat{\sigma}_k \ge cv_n$ on $\mathcal{H}_n$, direct normalization yields

$$\left[\mathbb{E}\{(G^*_{n,k}(t) - G^*_{n,k}(s))^2 \mid \mathcal{F}_n\}\right]^{1/2} \le CD_n|t-s|/v_n.$$

This argument avoids differentiating the estimated variance and retains the same observations used in the residual fit.

Conditionally, the process is Gaussian and is continuous on $\mathcal{V}_n$. The dyadic argument in Lemma S2.1 gives, on $\mathcal{H}_n$,

$$\mathbb{E}\{\omega_n(G^*_n) \mid \mathcal{F}_n\} \le CD_n\Delta_n\sqrt{L_n}/v_n.$$

Since $\overline{F}_1 \le \overline{F}_2^{1/2}$, $\mathbb{E}\{(1+\overline{F}_1)\overline{F}_2^{1/2}\} \le CK_n^2$. Conditional Markov's inequality and then unconditional Markov's inequality give

$$\begin{aligned}\mathbb{P}(r_n > n^{-1}) &\le \mathbb{P}(\mathcal{H}_n^c) + n\mathbb{E}(\mathbf{1}_{\mathcal{H}_n} r_n) \\ &\le Ce_n + Cn\epsilon_n^{-1} p^{3/2} K_n^2 h^{-3} \Delta_n \sqrt{L_n}/v_n \\ &\le C(e_n + n^{-1}).\end{aligned}$$

The final inequality follows directly from the chosen mesh and $L_n \le Cnp/h$. The same calculation shows $r_n = O_P(n^{-1})$ because $\mathbb{P}(\mathcal{H}_n^c) \to 0$. □

S2.4 Proof of Theorem 3

On $\mathcal{V}_n$, conditional on the observations,

$$\mathbb{E}\{G_n^*(s)G_n^*(t)^\intercal \mid \mathcal{F}_n\} = \widehat{Q}_n(s,t).$$

In particular, every conditional coordinate variance is one; centering the fitted contributions across subjects is not required for this identity. Proposition 2.1 of Chernozhukov et al. (2022), applied conditionally to the signed grid vectors, gives

$$\sup_z |\mathbb{P}(\|G_n^*\|_{\chi_n} \le z \mid \mathcal{F}_n) - \mathbb{P}(\|G_n\|_{\chi_n} \le z)| \le C(\Delta_{Q,n} L_n^2)^{1/2}.$$

The result permits singular covariance matrices, including the empirical bootstrap covariance. Discretization and Gaussian anti-concentration give

the samplewise bound

$$\sup_z |\mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) - F_G(z)| \le C\{(\Delta_{Q,n} L_n^2)^{1/2} + n^{-1}\} + r_n \quad \text{on } \mathcal{V}_n. \tag{S2.15}$$

Equations (S2.3) and (S2.14) prove the conditional approximation rate $O_P(\sqrt{e_n}\, L_n)$. Here $e_n \ge p^{3/2}\delta_n \ge \sqrt{L_n/n}$ implies $n^{-1} \lesssim \sqrt{e_n}\, L_n$, so the discretization term is absorbed in the stated rate.

We also need an unconditional approximation for the feasible statistic. On $\{\vartheta_n \le u\}$, where $0 < u \le 1/2$, $T_n^\circ/(1+u) \le T_n \le T_n^\circ/(1-u)$. Using (S2.3), (S2.13), and (S2.11) at deterministic thresholds gives

$$\sup_z |\mathbb{P}(T_n \le z) - F_G(z)| \le C\{\gamma_n + e_n/u + uL_n + n^{-1}\}.$$

Taking $u = (e_n/L_n)^{1/2}$ proves

$$\sup_z |\mathbb{P}(T_n \le z) - F_G(z)| \le C\{\gamma_n + (e_n L_n)^{1/2}\}. \tag{S2.16}$$

The triangle inequality now yields

$$\sup_z |\mathbb{P}(T_n \le z) - \mathbb{P}(S_n^* \le z \mid \mathcal{F}_n)| = O_P\{\gamma_n + (e_n L_n^2)^{1/2}\}.$$

All stochastic orders here are over the observations; the conditional distribution is explicitly evaluated given $\mathcal{F}_n$.

S2.5 Proof of Theorem 4

For $0 < u \le 1/2$, put $a_n(u) = C_0\{(uL_n^2)^{1/2} + n^{-1}\}$, with C_0 sufficiently large. Equations (S2.15), (S2.3), and (S2.14) imply the deterministic tail

bound

$$\mathbb{P}\left\{\sup_z |\mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) - F_G(z)| > a_n(u)\right\} \le C\{e_n/u + n^{-1}\}. \quad \text{(S2.17)}$$

In particular, this step does not substitute an O_P statement into an unconditional coverage probability.

Let $q_G(v) = \inf\{z : F_G(z) \ge v\}$ and define the conditional bootstrap quantile $\widehat{q}_v$ by the same generalized-inverse convention. On the event in (S2.17) where the supremum distance is at most $a = a_n(u)$,

$$q_G(v-a) \le \widehat{q}_v \le q_G(v+a), \qquad a < v < 1-a. \quad \text{(S2.18)}$$

For the lower inequality, if $z < q_G(v-a)$, then $F_G(z) < v-a$ and the conditional bootstrap distribution is less than v at z. For the upper inequality, continuity of F_G gives $F_G\{q_G(v+a)\} = v+a$, so the conditional bootstrap distribution is at least v there. These arguments require no strict monotonicity of either distribution function.

For fixed $\alpha \in (0,1)$, apply (S2.18) at $v = 1-\alpha$ and apply (S2.16) at the two resulting deterministic thresholds. The probability of the exceptional event in (S2.17) is added explicitly. This gives

$$|\mathbb{P}(T_n \le \widehat{q}_{1-\alpha}) - (1-\alpha)| \le C\{\gamma_n + (e_n L_n)^{1/2} + e_n/u + \sqrt{u}L_n + n^{-1}\}.$$

If the total bound is small, take $u = (e_n/L_n)^{2/3}$; otherwise the claimed bound follows by increasing its constant. Since $e_n L_n^2 \to 0$, this choice

is eventually at most $1/2$ and $a_n(u) < \min(\alpha, 1-\alpha)$. The last display becomes

$$|\mathbb{P}(T_n \le \widehat{q}_{1-\alpha}) - (1-\alpha)| \le C\{\gamma_n + (e_n L_n^2)^{1/3}\}.$$

Here $(e_n L_n)^{1/2}$ is absorbed by $(e_n L_n^2)^{1/3}$ for all sufficiently large n. On $\mathcal{V}_n$, the event on the left is exactly simultaneous coverage over all times and coordinates. The probability of $\mathcal{V}_n^c$ is at most Ce_n by (S2.3), and is absorbed by the same bound for any fixed convention for the bands on that event.

S3. Sampling frequency, dimension, and bandwidth

This section records consequences of the main results and sharper inputs when observation counts are comparable. They are supplementary to the linearization, approximation, and coverage argument in the main article.

S3.1 Approximation rates across sampling frequencies

Suppose $c_m m \le m_i \le C_m m$ for fixed positive constants c_m, C_m. Writing $m_{H,n} = \kappa_{1,n}^{-1}$, define the effective sample size by

$$N_n = \frac{n}{v_n^2} = \frac{n}{1 + (m_{H,n}h)^{-1}} \asymp n \min(1, mh). \tag{S3.1}$$

Thus the scale changes from the number of observations in local windows, nmh, to the number of independent subjects, n. To preserve this dependence in the approximation bounds, set

$$\delta_n^\dagger = \sqrt{\frac{L_n}{N_n}} + \frac{K_n n^{1/q} L_n^2}{N_n},$$
$$\rho_n^\dagger = p^{q/(2q-2)} \sqrt{\frac{L_n}{N_n}} + \frac{p n^{1/q} L_n^2}{N_n},$$
$$B_n^\dagger = v_n \max\{p, \sqrt{p}\, K_n L_n\},$$
$$e_n^\dagger = \frac{p K_n^2 n^{2/q} L_n^3}{N_n} + p^{3/2}(\delta_n^\dagger + h^s).$$

Let $\gamma_n^\dagger$ be defined by (3.2) with $B_n^\dagger, \delta_n^\dagger, \rho_n^\dagger$ in place of B_n, δ_n, ρ_n.

Proposition S3.1 (Sampling-dependent approximation bounds)**.** *Under Assumptions 1–5 and comparable counts, Theorems 1–4 hold with each rate replaced by its dagger version, under the corresponding rate conditions.*

Proof. Let $H_i(t) = m_i^{-1} \sum_j K_h(T_{ij} - t)$. Partition $[0,1]$ into intervals of length between $h/2$ and h. Each kernel window intersects at most six intervals. Bernstein's inequality for their binomial counts and a union bound give

$$\left\| \sup_t H_i(t) \right\|_{L^q} \le C \left\{ 1 + \frac{\log(1/h)}{m_i h} \right\} \le C L_n v_n^2.$$

For example, the maximum bin count exceeds $C\{m_i h + \log(1/h) + z\}$ with probability at most e^{-z}; integration proves the displayed moment bound. The envelope F_i in Assumption 4 is independent of these counts. Every

design or score entry is bounded by $CF_iH_i(t)$, so its supremum has L^q norm at most $CK_nL_nv_n^2$. The coordinate variance remains at most Cv_n^2. The score maximal inequalities in Lemma S1.2 therefore give $\delta_n^\dagger$. The spectral design and normalized Euclidean Taylor envelopes are at most $CpL_nv_n^2$ in L^q, using independence of $\sup_u \|\mathbf{X}_i(u)\|_2^2$ and the observation counts in the windows. Their matrix variance bounds remain $Cp^{q/(q-1)}v_n^2$. The matrix second-moment inequality thus gives $\rho_n^\dagger$.

The standardized contribution envelope is at most $C\sqrt{p}K_nL_nv_n$. Furthermore,

$$\Lambda_n \lesssim \frac{1+(mh)^{-3}}{\{1+(mh)^{-1}\}^2} \lesssim v_n^2,$$

so its coordinate fourth moments are at most $Cp^2v_n^2$. These give the Gaussian approximation input $B_n^\dagger$ and an empirical covariance bound of order

$$p\sqrt{L_n/N_n} + \frac{pK_n^2n^{2/q}L_n^3}{N_n}.$$

The first term is absorbed by $p^{3/2}\delta_n^\dagger$. The samplewise residual discrepancy bounds are unchanged, giving $e_n^\dagger$. The original increment bounds remain available for interpolation. Substitution into the approximation and quantile proofs establishes the proposition. □

In particular, the leading Gaussian term is

$$\left\{\frac{\max(p^2, pK_n^2L_n^2)L_n^5}{N_n}\right\}^{1/4},$$

and the bias term is $\sqrt{pN_nL_n}\,h^s$. The expressions separate the effects of observation frequency and dimension. Once mh is bounded away from zero, increasing the observation count no longer changes the polynomial order of N_n; dimension and smoothing still affect accuracy. This is a statement about the displayed sufficient bounds, not an optimality assertion.

Proof of Corollary 1. With $m_i = m$, (S3.1) gives $N_n = n/(1 + (mh)^{-1}) \asymp n \wedge (nmh)$ and $m_{H,n} = m$. Also $K_n \le C$, and $h \ge n^{-A}$ implies $L_n \asymp \log n$. The two balances in (3.8) therefore give

$$N_n \gtrsim n^{4/q} L_n^{3-4/q}, \qquad N_n^{3/4} h^s \lesssim L_n^{5/4}. \tag{S3.2}$$

In particular N_n grows at least polynomially, and the refined inputs satisfy

$$\delta_n^\dagger + \rho_n^\dagger \le C \left\{ \sqrt{\frac{L_n}{N_n}} + \frac{n^{1/q} L_n^2}{N_n} \right\}, \qquad B_n^\dagger \le C v_n L_n.$$

The first balance implies $\rho_n^\dagger \to 0$. Substitution into the first two terms of (3.2), using $n/v_n^2 = N_n$, gives

$$\left(\frac{(B_n^\dagger)^2 L_n^5}{n} \right)^{1/4} \le C \left(\frac{L_n^7}{N_n} \right)^{1/4},$$
$$\frac{B_n^\dagger L_n^{3/2-1/q}}{n^{1/2-1/q}} \le \frac{C n^{1/q} L_n^{5/2-1/q}}{\sqrt{N_n}}.$$

The stochastic linearization term is at most

$$\left(\frac{\sqrt{npL_n}\,\rho_n^\dagger \delta_n^\dagger}{v_n} \right)^{1/2} \le C (N_n L_n)^{1/4} \left\{ \sqrt{\frac{L_n}{N_n}} + \frac{n^{1/q} L_n^2}{N_n} \right\}$$
$$= C \left\{ \frac{L_n^{3/4}}{N_n^{1/4}} + \frac{n^{1/q} L_n^{9/4}}{N_n^{3/4}} \right\}.$$

Since $q \geq 4$ and $L_n, N_n \geq 1$ eventually, the first term is bounded by $C(L_n^7/N_n)^{1/4}$ and the second by $Cn^{1/q}L_n^{5/2-1/q}/\sqrt{N_n}$. The bias term is $O(\sqrt{N_n L_n}\, h^s)$. Finally,

$$\frac{n^{1/q}L_n^{5/2-1/q}/\sqrt{N_n}}{(L_n^7/N_n)^{1/4}} = \left(\frac{n^{4/q}L_n^{3-4/q}}{N_n}\right)^{1/4},$$

so the first balance in (S3.2) absorbs the finite-moment term. The second balance also gives

$$\sqrt{N_n L_n}\, h^s \lesssim \frac{L_n^{7/4}}{N_n^{1/4}}.$$

Thus $\gamma_n^\dagger \lesssim (L_n^7/N_n)^{1/4} \to 0$. Proposition S3.1 and the substitutions $N_n \asymp n \wedge (nmh)$, $L_n \asymp \log n$ prove (3.9).

To verify that these same conditions also suffice for band coverage, use the definition of $e_n^\dagger$ and fixed p, K_n to obtain

$$e_n^\dagger L_n^2 \leq C\left\{\frac{L_n^{5/2}}{\sqrt{N_n}} + \frac{n^{2/q}L_n^5}{N_n} + L_n^2 h^s\right\}.$$

Here the term $n^{1/q}L_n^4/N_n$ from $\delta_n^\dagger L_n^2$ is bounded by the displayed second term. By (S3.2), the three displayed terms are bounded, respectively, by constant multiples of

$$n^{-2/q}L_n^{1+2/q}, \qquad n^{-2/q}L_n^{2+4/q}, \qquad n^{-3/q}L_n^{1+3/q}.$$

They tend to zero because $L_n \asymp \log n$. The coverage conclusion now follows from Proposition S3.1. The width assertion (3.10) follows from the two bounds in Section S3.2, using fixed p, $\log(p/h) \asymp \log(1/h)$, and $m_{H,n} = m$.

For the stated example, $q = 8$, $r \geq 1$, and $h = n^{-2/5}$ give $n \wedge (nmh) \geq n^{3/5}$ for every $m \geq 1$. This exceeds $n^{1/2}(\log n)^{5/2}$ eventually. Since $s \geq 2$, also

$$\{n \wedge (nmh)\}^{3/4} h^s \leq n^{3/4-4/5} = n^{-1/20},$$

which verifies the second balance. □

S3.2 Covariance transitions and band width

Write $\lambda_n = (m_{H,n}h)^{-1}$. The exact covariance identity in Section S1.2 distinguishes identical from distinct observation indices of the same functional error. Its local part has order λ_n. The distinct-index component is $O(1)$ and has order one as $\kappa_{1,n} \to 0$; it is zero when all $m_i = 1$. Consequently $\lambda_n \to \infty$, $\lambda_n \asymp 1$, and $\lambda_n \to 0$ describe local, mixed, and functional covariance scales. No sampling regime needs to be selected to implement the bootstrap.

For clarity, a Gaussian process with the exact oracle covariance can be constructed as a sum of two independent auxiliary Gaussian processes. Using the weights $\mathbf{A}_h$ from Lemma S1.4, these are

$$W_d(t) = \int \mathbf{A}_h(t,u)\mathbf{C}(u,u)^{1/2}\sqrt{f(u)}\, dB(u),$$
$$W_o(t) = \int \mathbf{A}_h(t,u)W(u)f(u)\, du,$$

where B is vector white noise and W is an independent Gaussian process with covariance $\mathbf{C}(u, v)$. Multiplying these processes by $\sqrt{\kappa_{1,n}}$ and $\sqrt{1-\kappa_{1,n}}$, respectively, gives the covariance of the unstandardized oracle sum. This construction represents a covariance identity; the model still contains only $\varepsilon_i(t)$.

The uniform operator bounds on the covariance and the local weights imply that the standardized full covariance differs from the standardized local one by $O(\lambda_n^{-1})$ when $\lambda_n \to \infty$. When $\lambda_n \to 0$, it differs by $O(h + \lambda_n)$ from the correlation kernel of

$$\mathbf{\Omega}(t)^{-1} W(t), \qquad \mathbb{E}\{W(u)W(t)^{\intercal}\} = \mathbf{C}(u, t).$$

Here $\mathbf{\Omega}(t) = \mathbb{E}\{\mathbf{X}(t)\mathbf{X}(t)^{\intercal}\}$. Applying the same grid comparison as in the bootstrap proof shows that the corresponding Gaussian suprema have distribution errors at most $C\{L_n/\sqrt{\lambda_n} + n^{-1}\}$ and $C\{L_n\sqrt{h + \lambda_n} + n^{-1}\}$, respectively. The first bound requires $\lambda_n^{-1} L_n^2 \to 0$ for convergence and the second requires $(h + \lambda_n) L_n^2 \to 0$. These sufficient conditions make explicit why a pointwise covariance transition alone does not justify a simultaneous distributional limit when dimension grows.

Let $w_{n,k}(t) = \widehat{q}_{1-\alpha}\widehat{\sigma}_k(t)/\sqrt{n}$. Under the conditions of Theorem 4, there are constants $0 < c < C$ such that, with probability tending to one,

$$c\left\{\frac{1}{\sqrt{n}} + \sqrt{\frac{\log(1/h)}{nm_{H,n}h}}\right\} \le \inf_{t,k} w_{n,k}(t),$$

$$\sup_{t,k} w_{n,k}(t) \le C\left\{\sqrt{\frac{\log(2p)}{n}} + \sqrt{\frac{\log(p/h)}{nm_{H,n}h}}\right\}.$$

The same statement holds under the refined rate conditions in Proposition S3.1. To see the upper bound, set $Z(t) = \mathbf{\Omega}(t)^{-1}W(t)$ and $\mathbf{B}_h(t,u) = \mathbf{A}_h(t,u)\mathbf{\Omega}(u)f(u)$. The exact reproduction identity

$$\int_{\mathcal{T}} \mathbf{B}_h(t,u)\,du = \mathbf{I}_p$$

follows from the definition of $\mathbf{D}_h(t)$. The weights have uniformly bounded operator-norm L^1 norms, and their L^1 increments are at most $C|t-s|/h$. The increment covariance of Z has operator norm at most $C|t-s|$. For $|t-s| \le h$, integrate $\{\mathbf{B}_h(t,u)-\mathbf{B}_h(s,u)\}\{Z(u)-Z(s)\}$ over the union of the kernel windows. Its coordinate L^2 norm is at most $C|t-s|/\sqrt{h} \le C\sqrt{|t-s|}$. For $|t-s| > h$, use $\max_{k\le p}\|W_{o,k}(t)-Z_k(t)\|_{L^2} \le C\sqrt{h}$. Thus the functional Gaussian component has coordinate increments bounded in mean square by $C|t-s|$ throughout $\mathcal{T}$. The local component has coordinate L^2 increments at most $Ch^{-1/2}\min\{1,|t-s|/h\}$. Gaussian entropy bounds over p coordinates give expected suprema at most $C\sqrt{\log(2p)}$ and $Ch^{-1/2}\sqrt{\log(p/h)}$. Gaussian concentration transfers these bounds to fixed quantiles. For the

lower bound, select one coordinate and $\asymp h^{-1}$ points with disjoint kernel windows. The local white-noise component at these points is independent, with variances bounded below by c/h. Condition on the independent functional Gaussian component. For each local normal variable, the probability of lying in an interval of a given length is maximized when that interval is centered. Hence, with $J \asymp h^{-1}$ and every $x > 0$, the probability that all selected sums have absolute value at most x is at most $\{2\Phi(x/\sqrt{c\lambda_n}) - 1\}^J$. This gives a fixed-quantile lower bound of order $\sqrt{\lambda_n \log(1/h)}$. A single nondegenerate coordinate gives a lower bound of order v_n, which also supplies the stated constant-order term. Uniform variance consistency and the bootstrap quantile bracket transfer both bounds to $w_{n,k}$.

When p is bounded these two orders agree. For growing p, the displayed lower bound uses a single coordinate and is deliberately not sharpened in the dimension. The sparse-to-dense variance scale and the cost of simultaneous coverage over many functions are therefore distinct.

S3.3 Bandwidth choice and implementation

For increasing p, admissibility is determined by $\gamma_n \to 0$ and $e_n L_n^2 \to 0$, or their refined versions. These conditions include the undersmoothing requirement $\sqrt{npL_n}\, h^s / v_n \to 0$. They are nonempty for growing VCMs:

with $r = 1$, $q = 8$, $m_i = 1$, $h = n^{-1/4}$, and $p \asymp n^a$ for any $0 < a < 1/4$, the general bounds vanish even using $K_n \lesssim p^{1/8}$. Indeed,

$$\gamma_n = O\{\sqrt{p}\, n^{-1/8}\sqrt{\log n}\}, \qquad e_n = O\{p^{3/2} n^{-3/8}\sqrt{\log n}\},$$

which verify both conditions throughout this range. The dimension restriction comes from the displayed sufficient bounds and is not claimed to be sharp.

For the numerical comparisons, whose dimensions are specified in advance, we retain the local linear bandwidth

$$h = \{1 + \log n\}^{-1/2} \max\{n^{-1/4}, (n m_{H,n})^{-1/5}\}. \tag{S3.3}$$

For bounded p, this rule satisfies (3.7). Indeed $L_n \asymp \log n$. If $n^{-1/4}$ is the larger term, use $v_n \geq 1$ to obtain normalized bias $O\{(\log n)^{-1/2}\}$. If $(n m_{H,n})^{-1/5}$ is larger, use $v_n \geq (m_{H,n} h)^{-1/2}$ to obtain $O\{(\log n)^{-3/4}\}$. This formula specifies the numerical procedure. It is not a universal bandwidth rule for arbitrary growing dimension; the dimension-dependent conditions above must be checked for such sequences. All fits, residuals, and multiplier contributions use the selected common bandwidth. An independent pilot may select among bandwidths for which the displayed errors vanish uniformly: conditioning on that pilot and integrating the coverage bound proves the same conclusion for the inference sample.

For implementation on a grid of mesh Δ, the increment bound used in Section S2 gives, on the good design and variance event, a conditional expected interpolation error bounded by $C\mathcal{L}_n\Delta\sqrt{L_n + \log(1/\Delta)}$, where

$$\mathbb{E}\{\mathcal{L}_n \, \mathbf{1}_{\text{good}}\} \leq Cp^{3/2}K_n^2/(v_n h^3).$$

It is enough to use the theoretical grid constructed there, whose additional probability error is $O(n^{-1})$. More generally, a grid error term tends to zero whenever

$$\frac{p^{3/2}K_n^2\Delta}{v_n h^3}\sqrt{L_n\{L_n + \log(1/\Delta)\}} \to 0.$$

To verify this assertion, use conditional Markov's inequality twice and Gaussian anti-concentration. Balancing the two Markov errors and the anti-concentration shift yields the cube root of this displayed quantity, which can be added to the coverage bound. This sufficient condition does not certify a fixed grid such as 201 points. The simulations report coverage on their stated grid.

With B independent multiplier draws, the conditional empirical cdf has uniform error at most $\epsilon_B = \sqrt{\log(2/\eta)/(2B)}$ with conditional probability at least $1 - \eta$, by the Dvoretzky–Kiefer–Wolfowitz inequality (Massart, 1990). Adding this cdf error to the same quantile bracket adds at most $C\epsilon_B + \eta$ to the coverage bound, including the multiplier randomness. This separates statistical, grid, and simulation errors.

S4. Proof of the mean-function corollary

We use the assumptions, rates, and mean-model statistics from Section 3.5. All bounds below are uniform in t, k; r, q are fixed, and constants are uniform in n, p, h and the counts. The argument specializes the subject-level linearization and covariance bounds before applying any dimension-dependent coordinate-mixing bound.

Proof of Corollary 2. Let $H(t) = \mathbb{E}\widehat{H}(t)$ and write the oracle contributions as

$$w_h(t,u) = \mathbf{e}_0^\top H(t)^{-1}\boldsymbol{\theta}_r\left(\frac{u-t}{h}\right)K_h(u-t),$$

$$\Psi_{ik}(t) = \frac{1}{m_i}\sum_j w_h(t,T_{ij})\varepsilon_{i,k}(T_{ij}).$$

The population matrix H has fixed size $r+1$ and is uniformly invertible by Lemma S1.1, applied to scalar predictor one. The coordinatewise covariance calculation in Lemma S1.4 gives $cv_n^2 \le \tau_k^2(t) \le Cv_n^2$. Only the covariance entries involving the specified coordinates occur in these calculations; an operator-norm bound on a $p \times p$ covariance matrix is unnecessary.

Put

$$D_n = \sup_t \|\widehat{H}(t) - H(t)\|_{\mathrm{op}},$$

$$U_n = \sup_{t,k}\left\|\frac{1}{n}\sum_i \frac{1}{m_i}\sum_j K_h(T_{ij}-t)\boldsymbol{\theta}_r\left(\frac{T_{ij}-t}{h}\right)\varepsilon_{i,k}(T_{ij})\right\|_2.$$

For each design entry, mutual independence of the observation times gives variance at most $C\kappa_{1,n}/(nh)$ and maximal centered summand at most $C/(nh)$. Use a design grid of mesh at most $n^{-8}h^{10}$, whose log cardinality is $O\{\log(n/h)\}$, independently of p. Each entry of $\widehat{H} - H$ has deterministic Lipschitz constant at most Ch^{-2}. Bernstein's inequality, integration of its tail, and interpolation therefore yield

$$\mathbb{E}D_n^2 \leq Ca_n^2.$$

For the scores, the average coordinate second moment is at most Cv_n^2, and the common subject envelope has L^q norm at most CK_n/h. The maximal inequality and grid interpolation used in Lemma S1.2 give

$$\mathbb{E}U_n^2 \leq C\delta_n^2.$$

Moreover, $a_n \leq \delta_n$, since $\log(n/h) \leq L_n$, $K_n \geq 1$, and the finite-moment term in δ_n dominates $L_n/(nh)$.

On the good-design event $\mathcal{E}_n = \{D_n \leq c_0\}$, inversion is Lipschitz and the Taylor remainder is uniformly bounded by Ch^s. Consequently,

$$\|\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|_{\infty,\mathcal{T}} \leq C(U_n + h^s), \qquad \|\mathbf{R}_n\|_{\infty,\mathcal{T}} \leq C(D_nU_n + h^s).$$

Since $\mathbb{P}(\mathcal{E}_n^c) \le Ca_n^2$ and $\mathbb{E}(D_nU_n) \le Ca_n\delta_n$, these inequalities prove the linear representation and, for every $u > 0$, the quantitative transfer

$$\mathbb{P}\left\{\sup_{t,k}\frac{\sqrt{n}|R_{nk}(t)|}{\tau_k(t)} > C\frac{\sqrt{n}h^s}{v_n} + u\right\} \le Ca_n^2 + \frac{C\sqrt{n}\,a_n\delta_n}{v_n u}.$$

The smoothing shift is thus deterministic on an event with a controlled failure probability, as needed for the coverage argument.

Let $M_{ik}(t) = \Psi_{ik}(t)/\tau_k(t)$. The repeated-observation-index moment calculation gives

$$\max_i \left\|\sup_{t,k}|M_{ik}(t)|\right\|_{L^q} \le \frac{CK_n}{hv_n},$$
$$\sup_{t,k}\frac{1}{n}\sum_i \mathbb{E}|M_{ik}(t)|^4 \le C\Lambda_n.$$

Because $\kappa_{3,n} \le \kappa_{1,n}$ and $h \le 1$, $\Lambda_n \le C/(h^2v_n^2) \le CK_n^2/(h^2v_n^2)$. Also $h^2v_n^2 \le 3/4$ and $K_n \ge 1$. Theorem 2.5 of Chernozhukov et al. (2022) therefore applies to the signed grid with moment scale $K_n/(hv_n)$, giving the first two terms of γ_n^μ. The grid in Lemma S2.1 has log cardinality $O(L_n)$ because $K_n \le Cp^{1/q}$. The same fixed-coordinate increment bounds give a continuous Gaussian version, an $O(n^{-1})$ continuum error, and anti-concentration at most $Cu\sqrt{L_n}$. Balancing this last error with the transfer probability gives

$$C\left(\frac{\sqrt{nL_n}\,a_n\delta_n}{v_n}\right)^{1/2} + \frac{C\sqrt{nL_n}\,h^s}{v_n}.$$

The stochastic term is at least a constant times $\sqrt{L_n a_n}$, so it absorbs a_n^2 when $a_n \to 0$. The first Gaussian term absorbs n^{-1}. This proves (3.13).

For oracle covariance products, the average coordinate variance is at most $C\Lambda_n$. The L^2 norm of their envelope maximized over subjects is at most $Cn^{2/q}K_n^2/(h^2 v_n^2)$. Applying the product maximal inequality over coordinate and grid pairs, followed by interpolation, gives expected covariance error at most

$$C\left\{\sqrt{\frac{\Lambda_n L_n}{n}} + \frac{K_n^2 L_n}{n^{1-2/q}h^2 v_n^2}\right\}.$$

The empirical quadratic averages of the oracle scores and local kernel masses are bounded by fixed constants outside an event with probability controlled by this same bound. On this event and $\mathcal{E}_n$, samplewise Cauchy–Schwarz gives

$$\sup_{t,k}\left[\frac{1}{n}\sum_i\left\{\frac{\widehat{\Psi}_{ik}(t)-\Psi_{ik}(t)}{\tau_k(t)}\right\}^2\right]^{1/2} \le C(D_n + U_n + h^s).$$

Here the exact residual identity is $\widehat{\varepsilon}_{ij,k} - \varepsilon_{i,k}(T_{ij}) = -\{\widehat{\mu}_k(T_{ij}) - \mu_k(T_{ij})\}$. The truncated first and second moments of the fitted-score discrepancy are $O(\delta_n + h^s)$ when $e_n^\mu \to 0$, using $a_n \le \delta_n$. All exceptional probabilities are $O(e_n^\mu)$ as well. The covariance-product identity and variance normalization therefore imply

$$\mathbb{P}(\Delta_n^\mu > u) \le \frac{Ce_n^\mu}{u}, \qquad 0 < u \le 1/2, \tag{S4.1}$$

where Δ_n^μ is the sum of relative variance error and standardized fitted covariance error, set to infinity on invalid samples. This also controls failure of fitted variance positivity.

The conditional discretization proof in Lemma S2.3 applies with fixed matrix size $r+1$. On its good event the expected random Lipschitz factor is at most $CK_n^2/(v_n h^3)$, and the event's complement has probability $O(e_n^\mu)$. For the conditional interpolation error probability r_n, the same grid and two applications of Markov's inequality give

$$\mathbb{P}(r_n > n^{-1}) \le C(e_n^\mu + n^{-1}), \qquad r_n = O_P(n^{-1}).$$

Conditional Gaussian comparison and (S4.1) now prove the first bootstrap assertion. The second follows from the same studentization bracket as in Theorem 3; the unconditional feasible-statistic error is at most $C\{\gamma_n^\mu + (e_n^\mu L_n)^{1/2}\}$. The discretization terms are absorbed because $e_n^\mu \ge \delta_n \ge v_n\sqrt{L_n/n}$.

For coverage, (S4.1) and the conditional grid bound imply, for $0 < u \le 1/2$,

$$\mathbb{P}\Big\{\sup_z \big|\mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) - \mathbb{P}(S_{G,n} \le z)\big| > C(\sqrt{u}L_n + n^{-1})\Big\} \le C(e_n^\mu/u + n^{-1}).$$

Bracket the random bootstrap quantile between the two Gaussian quantiles as in (S2.18), and add this bad-event probability. The resulting unconditional coverage error is at most

$$C\{\gamma_n^\mu + (e_n^\mu L_n)^{1/2} + e_n^\mu/u + \sqrt{u}L_n + n^{-1}\}.$$

Taking $u = (e_n^\mu/L_n)^{2/3}$ proves (3.14) when the bound is small; otherwise the trivial bound suffices after increasing C. The probability of invalid design or variances is $O(e_n^\mu)$ and is included explicitly. Thus the conclusion controls unconditional coverage, including estimation, residual fitting, studentization, and the random critical value. □

S4.1 A high-dimensional example with polynomial tails

Let V_i be a centered Student t_ν variable, standardized to have variance one, with $\nu > 8$. Independently, let $R_{ik,0}, R_{ik,1}$ be independent Rademacher variables, and set

$$\varepsilon_{i,k}(t) = V_i\{R_{ik,0} + R_{ik,1}\sin(2\pi t)\}.$$

The errors have polynomial tails and $\sup_{t,k}|\varepsilon_{i,k}(t)| \le 2|V_i|$, so their L^8 envelope is bounded independently of p. Their covariance entries are $C_{kl}(u,t) = \mathbf{1}_{\{k=l\}}\{1 + \sin(2\pi u)\sin(2\pi t)\}$, which are uniformly Lipschitz, with coordinate variances at least one. The shared scale V_i permits dependence across coordinates.

Take $q = 8$, local linear fits, $m_i = 1$, and $h = n^{-1/4}$. Then $v_n \asymp n^{1/8}$ and $K_n = O(1)$. The displayed rates give

$$\gamma_n^{\mu} = O\{n^{-3/16} L_n^{5/4} + n^{-1/4} L_n^{11/8} + n^{-1/8} L_n^{1/2}\},$$
$$e_n^{\mu} L_n^2 = O\{n^{-3/8} L_n^{5/2} + n^{-1/2} L_n^3\}.$$

If $\log p = o(n^{3/20})$, both vanish, and the first covariance term in $(e_n^{\mu} L_n^2)^{1/3}$ dominates all the displayed contributions to coverage. The resulting error is $O(n^{-1/8} L_n^{5/6})$. The undersmoothing condition is also explicit: $\sqrt{nL_n}\, h^2 / v_n \asymp n^{-1/8}\sqrt{L_n} \to 0$. Thus $p = \lceil \exp(n^{\delta}) \rceil$ is admissible for every fixed $0 < \delta < 3/20$ under polynomial moments, even with one observation per subject. This is a sufficient example, not an optimal dimension range.

S5. Reproducibility and additional numerical results

S5.1 Simulation generating mechanism

The factorial design uses $n \in \{200, 400\}$ subjects and $m \in \{5, 20, 80\}$ observations per subject, with 1,000 datasets in each configuration. All observation times are independent Uniform$(0, 1)$. Set $b(t) = (1, \sin(2\pi t), \cos(2\pi t))^{\intercal}$ and define

$$\mathbf{X}_i(t) = \{1, A_{i1}^{\intercal} b(t)/\sqrt{2}, \ldots, A_{i,p-1}^{\intercal} b(t)/\sqrt{2}\}^{\intercal},$$

where all entries of the A_{ik} are independent standard normals. Independently, generate

$$\varepsilon_i(t) = Z_{i0} + \tfrac{1}{2} Z_{i1} \sin(2\pi t) + \tfrac{1}{2} Z_{i2} \cos(2\pi t).$$

The Z variables are independent draws from one of three laws: $N(0,1)$, Laplace$(0, 1/\sqrt{2})$, or $(\chi^2_5 - 5)/\sqrt{10}$. Here the Laplace parameter is its scale; the chi-square law is centered and standardized by its mean 5 and standard deviation $\sqrt{10}$. All three have mean zero, variance one, and finite moments of every fixed order. Responses are generated as $Y_{ij} = \mathbf{X}_i(T_{ij})^\intercal \boldsymbol{\beta}(T_{ij}) + \varepsilon_i(T_{ij})$, using this single functional error. Thus $\boldsymbol{\Omega}(t) = \mathbf{I}_p$ and $\mathbf{C}(t,t) = 1.25\mathbf{I}_p$ for every law.

For $p = 2$, take $\beta_1(t) = 0.5 + 0.5t$ and $\beta_2(t) = 0.5 + 0.4\sin\{2\pi(t - 0.5)\}$, and target β_2. For $p = 5$, add $\beta_3(t) = 0.3\cos(2\pi t)$, $\beta_4(t) = 0.5t^2$, and $\beta_5(t) = 0.4e^{t-1}$, and target all five functions jointly. The intercept-only mean comparison uses the same error process and the mean curve specified below. Local linear fits use the Epanechnikov kernel and $B = 1{,}999$ Gaussian multiplier draws. All comparisons evaluate coverage and width on the same 201 equally spaced points in $[0.15, 0.85]$.

S5.2 Computation and comparator formulas

Epanechnikov-weighted local moments are evaluated by exact polynomial prefix sums. Observation times are not binned, and residuals are fitted at every actual observation time, throughout the observation domain. Local Gram eigenvalues and variance positivity are checked; no ridge term or silent regularization is applied. A failed fit is counted as noncoverage. All methods within a comparison use the same dataset and target event.

The simulation bandwidth is (S3.3), with multiplicative constant one. It is determined by n and the counts, without using observed coverage or responses. The estimate, residuals, and multiplier covariance use that common value. This is a controlled comparison at specified tuning parameters, not a claim about the relative performance of all possible tuning rules.

For CLZ, put $N = \sum_i m_i$ and use the pooled local linear fit and its residuals. We use the same estimation bandwidth for both methods and for all nuisance kernel estimates; the original CLZ cross-validation choices are not imposed on either estimator. With equal counts and this common bandwidth, the two point estimates agree. Following equations (3.2), (3.5),

and (3.6) of Cao, Liu and Zhou (2018), compute

$$\widehat{f}(t) = N^{-1} \sum_{ij} K_h(T_{ij} - t),$$

$$\widehat{\Omega}(t) = N^{-1} \sum_{ij} K_h(T_{ij} - t) \mathbf{X}_{ij} \mathbf{X}_{ij}^{\intercal},$$

$$\widehat{\Lambda}(t) = N^{-1} \sum_{ij} K_h(T_{ij} - t) \mathbf{X}_{ij} \mathbf{X}_{ij}^{\intercal} \widehat{\varepsilon}_{ij}^2.$$

The pointwise standard error for coefficient k is

$$\left[\frac{\nu_K}{N h \widehat{f}(t)} \{ \widehat{\Omega}(t)^{-1} \widehat{\Lambda}(t) \widehat{\Omega}(t)^{-1} \}_{kk} \right]^{1/2}, \qquad \nu_K = \int K^2 = 3/5.$$

These formulas reproduce the stated raw kernel-moment normalization for the uniform-design experiments. They are not presented as density-normalized conditional-moment estimates for an arbitrary nonuniform design. The normalization uses the total observation count N, not the number of subjects.

Their Section 3 critical-value simulation generates fresh independent $U_1, \ldots, U_N \sim \text{Uniform}(0,1)$ and $g_1, \ldots, g_N \sim N(0,1)$, and records

$$\max_{t \in \mathcal{G}} \frac{\left| \sum_{a=1}^{N} g_a K\{(U_a - t)/h\} \right|}{\sqrt{N h \nu_K}}.$$

We use 20,000 independent draws for each (n, m) combination. The empirical 0.95 quantile gives a single-coefficient band; the 0.99 quantile gives each of the five marginal bands in the Bonferroni comparison. Sorted uniform times are generated exactly from normalized exponential spacings, and the

polynomial kernel sums are evaluated exactly. The calibration maxima are retained. CLZ's sparse-sampling theory does not provide a dense-sampling guarantee.

For the mean comparison, set $X_i(t) \equiv 1$ and

$$\mu(t) = \sin(\pi t) + t + 0.25\{\cos(2\pi t) + \sin(2\pi t)\},$$

a curve also used by Lou and Wu (2026). Retain the same functional error $\varepsilon_i(t)$ generated from Gaussian, Laplace, or adjusted chi-square innovations as in the main experiments. For the Cai–Hu method, let $B_J(t)$ be the cubic B-spline vector with J equally spaced interior knots and repeated endpoint knots. The pooled fit minimizes squared residuals over all N observations. We use the selection rule in equation (5.1) of the public author version, `https://arxiv.org/pdf/2401.17646`, of Cai and Hu (2025):

$$\mathrm{BIC}(J) = \log\left[N^{-1}\sum_{ij}\{Y_{ij} - \widehat{\mu}_J(T_{ij})\}^2\right] + J\log(N)/N,$$

with integer candidates from $\lceil 0.5N^{1/6}\rceil$ to $\lfloor 2N^{1/4}\rfloor$. For the selected J, put

$$A = \sum_{ij} B_J(T_{ij})B_J(T_{ij})^{\mathsf{T}}, \qquad s_i = \sum_j B_J(T_{ij})\widehat{\varepsilon}_{ij}.$$

The covariance of the spline coefficient estimate is

$$\widehat{V}_\theta = A^{-1}\left(\sum_i s_i s_i^{\mathsf{T}}\right)A^{-1}.$$

The standard error is $\{B_J(t)^\intercal \widehat{V}_\theta B_J(t)\}^{1/2}$. Gaussian coefficient draws from this covariance yield the studentized grid maximum. This comparison uses the same $B = 1{,}999$ and evaluation grid as the proposed method.

Computational checks compare the prefix-sum local fits and subject contributions with direct weighted matrix calculations; verify kernel covariance by quadrature; check both kernel-moment normalizations against direct formulas; verify the finite-observation calibration sums; and check equal-count weighting and the spline subject sandwich. Generator checks verify the single functional-error model and the innovations' centering, variance, and skewness. Further checks verify the functional-error covariance identity by independent quadrature and the rate-consolidation identities algebraically. A fresh rerun of replication zero in all 54 configurations reproduces every inferential result field.

S5.3 Random streams and retained records

There are 54 configurations: three inferential targets, three innovation laws, two subject sample sizes, and three sampling frequencies. Each has 1,000 replications. The configuration's stored `seed_id`, denoted by j, and replication r specify the NumPy seed sequence $[20261015, 100000(j+1) + r]$.

Three independent child streams generate data and the two methods' multiplier draws. CLZ calibration uses the independent entropy $[713209, n, m]$. No configuration or replication is omitted according to its outcome.

The supplied `replications.jsonl` records the error-model version, configuration, replication, bandwidth, critical values, coverage indicators, widths, minimum local eigenvalue, and failures. The summary contains Monte Carlo standard errors, Wilson intervals, and paired differences of coverage indicators. The complete experiment contains 54,000 datasets, with both methods fitted to every dataset. The six Gaussian joint cells use seed identifiers 2006, 2007, 2008, 2010, 2011, and 2012 and were recomputed independently with 1,000 datasets each. The current revision retains 36,000 unchanged Gaussian and Laplace records with source provenance and computes all 18,000 adjusted χ^2_5 records afresh. All previous Gaussian joint records are archived; comparisons between independent cohorts and the $m = 20$ versus 80 CP differences, including Monte Carlo uncertainty, are retained with the results. The runner and table generator check the error-model version and reject incomplete or duplicate records. Package versions and execution commands accompany the code.

For each of the six Gaussian joint configurations, an additional check computes weighted least-squares fits and subject contributions directly at

all 201 grid points for all five coefficients, using replication zero. Residual fits are checked at 17 actual observation times, including both extremes. All 1,999 bootstrap maxima agree with the prefix-sum implementation to within 2×10^{-10}. This check retains the actual counts and bandwidth of each configuration.

S5.4 Additional simulation results

Table S1 reports coverage and width for β_2 in the two-coefficient model. It complements the joint five-function comparison in main-text Table 1; both tables retain every sampling frequency and error distribution in the factorial design.

Table S2 reports the scalar-mean comparison, in which both procedures retain subject dependence. The growing-dimensional mean result is Corollary 2; its proof is in Section S4.

Table S1: Coverage probability (CP) and average width (AW) of nominal 95% bands for β_2 in the two-coefficient model. Each row uses 1,000 datasets; $m = 5, 20, 80$ denote sparse, mild, and dense sampling.

			Proposed		CLZ	
Innovations	n	m	CP	AW	CP	AW
Gaussian	200	5	0.916	0.599	0.735	0.485
	200	20	0.925	0.460	0.326	0.244
	200	80	0.930	0.411	0.052	0.121
	400	5	0.908	0.473	0.763	0.395
	400	20	0.925	0.347	0.389	0.197
	400	80	0.925	0.303	0.076	0.099
Laplace	200	5	0.910	0.594	0.727	0.481
	200	20	0.933	0.452	0.341	0.240
	200	80	0.936	0.409	0.053	0.121
	400	5	0.912	0.470	0.755	0.393
	400	20	0.927	0.345	0.396	0.197
	400	80	0.929	0.302	0.083	0.099
Adjusted χ^2_5	200	5	0.908	0.593	0.733	0.479
	200	20	0.924	0.457	0.353	0.243
	200	80	0.925	0.411	0.059	0.121
	400	5	0.912	0.472	0.773	0.395
	400	20	0.948	0.347	0.397	0.197
	400	80	0.922	0.302	0.083	0.099

Table S2: Coverage probability (CP) and average width (AW) of nominal 95% bands for the mean function. Each row uses 1,000 datasets; $m = 5, 20, 80$ denote sparse, mild, and dense sampling.

			Proposed		Cai–Hu	
Innovations	n	m	CP	AW	CP	AW
Gaussian	200	5	0.937	0.609	0.939	0.534
	200	20	0.941	0.458	0.945	0.427
	200	80	0.948	0.400	0.954	0.396
	400	5	0.942	0.479	0.938	0.379
	400	20	0.950	0.345	0.947	0.313
	400	80	0.938	0.293	0.943	0.281
Laplace	200	5	0.946	0.608	0.947	0.533
	200	20	0.941	0.457	0.952	0.427
	200	80	0.954	0.398	0.958	0.394
	400	5	0.936	0.477	0.936	0.378
	400	20	0.939	0.344	0.938	0.313
	400	80	0.932	0.293	0.941	0.280
Adjusted χ^2_5	200	5	0.928	0.609	0.951	0.534
	200	20	0.925	0.455	0.937	0.425
	200	80	0.938	0.401	0.944	0.397
	400	5	0.919	0.477	0.932	0.379
	400	20	0.945	0.344	0.947	0.312
	400	80	0.941	0.293	0.941	0.281

S5.5 Multivariate mean experiment

The additional mean experiment uses the fixed factorial design $n \in \{200, 400\}$, $m_i = m \in \{5, 20, 80\}$, and $p \in \{50, 100\}$. With independent Uniform$(0, 1)$ observation times, set

$$\begin{aligned}
\mu_k(t) &= \sin(\pi t) + \tfrac{1}{2}\cos(\theta_k)t + \tfrac{1}{4}\sin(\theta_k)t^2, \\
\theta_k &= 2\pi\{(k-1) \bmod 20\}/20, \\
\varepsilon_{i,k}(t) &= \frac{1}{\sqrt{2}}\big[U_{i0} + Z_{ik0} + \tfrac{1}{2}(U_{i1} + Z_{ik1})\sin(2\pi t) \\
&\qquad + \tfrac{1}{2}(U_{i2} + Z_{ik2})\cos(2\pi t)\big].
\end{aligned}$$

All U and Z variables are mutually independent innovations from the same selected law: $N(0, 1)$, Laplace$(0, 1/\sqrt{2})$, or $(\chi^2_5 - 5)/\sqrt{10}$. The U terms are shared by all coordinates of a subject. Each error coordinate has variance 1.25, and distinct coordinates have contemporaneous correlation 1/2. Thus the Gaussian case is jointly Gaussian, while the other cases retain the selected innovations' heavier tails or skewness. All satisfy the polynomial-moment assumptions of Corollary 2. There is one vector-valued functional error and no added observation noise.

Each $(n, m, \text{innovation law})$ uses 1,000 independent datasets, giving 18,000 independent datasets across the 18 sampling and innovation groups. The two dimensions use nested coordinate prefixes and the same datasets and multipliers, giving 36,000 dimensionwise evaluations in 36 cells. Every

configuration is reported regardless of its coverage. With 1,000 replications, the MCSE of CP near 0.95 is approximately 0.007. Every band uses 999 Gaussian multiplier draws, with one multiplier shared by all coordinates of a subject. Coverage requires every coordinate at all 51 equally spaced evaluation points on $[0, 1]$, including the endpoints. These are grid-coverage measurements. The common bandwidth is (S3.3); it does not use responses or coverage outcomes. All observations of subject i evaluate the same error curve $\varepsilon_{i,k}$ defined above.

The implementation shares the scalar local design and fits response coordinates in blocks of at most 128. Exact polynomial prefix sums evaluate residual fits at every actual observation time without forming an observations-by-observations smoothing matrix. Bootstrap maxima are accumulated over all blocks and grid points before taking the empirical 0.95 quantile. No coordinate subsampling, marginal critical values, covariance regularization, or bias correction is used. Independent checks compare the blockwise estimates, residuals, joint bootstrap maxima, critical values, and widths with coordinatewise applications of the scalar code, including endpoints, unequal counts, and shuffled observations. For $n = 400$, $m = 20$, and $p = 50$, complete Gaussian and adjusted chi-square replication-zero

calculations also agree with the original scalar implementation to within 2×10^{-8} for all 999 bootstrap maxima.

The seed sequence is $[20261017, n, m, \ell, r]$, where $\ell = 0, 1, 2$ identifies the Gaussian, Laplace, and adjusted chi-square laws, respectively, and $r = 0, \ldots, 999$. Four child streams generate times, common innovations, coordinate-specific innovations, and multipliers. The records in `results/highd-mean/` retain coverage, width, critical values, design and variance diagnostics, conditional smoothing bias divided by the fitted standard error, failures, and time. They also retain code hashes and runtime versions. The fixed design and production-code hashes accompany the results. CP standard errors and Wilson intervals are computed from the 1,000 independent replications in each cell. The new implementation uses Python 3.12.14 and NumPy 2.5.4 with Accelerate BLAS. The mean production run uses six processes alongside four processes for the comparison experiment, with thread-limit environment variables set to one. The retained per-cell time is response-block fitting and calibration time at that dimension; shared design and multiplier setup and initial coefficient generation are excluded. End-to-end times are retained separately. This experiment is independent

of the 54,000-dataset factorial comparison. All multivariate-mean evaluations were recomputed on the 51-point grid; timing records therefore refer to this run. Execution metadata are supplied with the results.

The fixed design contains 18,000 independent subject datasets and 36,000 dimensionwise evaluations. The two dimensions use nested coordinates and the same subject multipliers. The production run computed all 36,000 evaluations in 28.2 minutes using 6 workers.

Replication zero was independently reproduced in all 36 cells. Refining those datasets from 51 to 101 grid points changed 0 coverage judgments; the largest relative critical-value increase was 0.780%. This selected-dataset check does not establish continuum coverage.

S5.6 PBC preprocessing and sensitivity

The raw CSV and its SHA-256 digest are supplied with a provenance record. It contains 1,945 visits from 312 patients. For each analysis horizon, we first restrict the visit days, then remove rows missing any variable needed in that model; bilirubin and alkaline phosphatase must be positive before logarithms are taken. Counts are recomputed after filtering, and weights are $1/m_i$ for those retained counts. Standardization uses observation-level empirical means and standard deviations with divisor N. The actual retained

rows, centering and scaling constants, local support counts, estimates, standard errors, and critical values are saved. Thus the exploratory fitted target and its preprocessing are reproducible.

For the 3,000-day horizon, the mean analysis has $N = 1{,}790$ and $m_H = 3.6088$; the six-coefficient analysis has $N = 1{,}741$ and $m_H = 3.4890$. The latter uses a 201-point grid from 450 to 2,550 days. At the primary bandwidth, the nominal positive exclusions for edema are days 733.5–1,311, 1,510.5–1,878, 1,951.5–2,004, and 2,287.5–2,550. For prothrombin time they are days 450–576 and 1,069.5–2,140.5. The albumin exclusions are negative on days 544.5–607.5 and 649.5–775.5. These endpoints refer only to evaluated grid points; they are not estimated change points. The treatment band contains zero throughout, and the log-ALP band is positive throughout.

At bandwidth factor 0.8, the critical value is 3.582 and the albumin band contains zero on the entire grid. At factor 1.2, the critical value is 3.358 and albumin has a negative exclusion extending from day 450 to day 775.5. Treatment and log-ALP retain their respective descriptions across all three factors. The common-random-number seed for the 3,000-day six-coefficient multiplier draws is 530106.

Restricting the horizon to day 2,000 gives 1,518 visits for the mean fit and 1,484 complete visits for the regression, both from 312 patients.

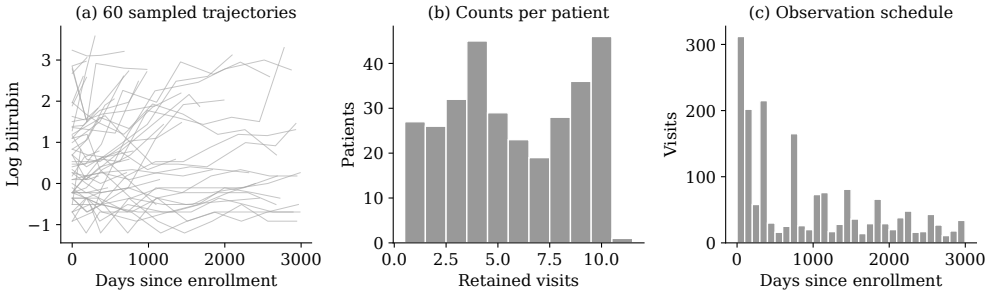


Figure S1: PBC trajectories for 60 patients selected with a fixed seed, retained counts, and visit times up to day 3,000. Baseline and scheduled visits are visible.

After rescaling by 2,000, the regression bandwidth is 0.09598 (about 192 days), $m_H = 3.3330$, and the critical value is 3.576. On the resulting evaluation grid, days 300–1,700, the treatment band again contains zero and the log-ALP band remains positive. The shorter biomarker exclusions vary. Changing the horizon also changes the empirical scaling constants and the composition of observed visits; it is a descriptive sensitivity analysis, not a correction for informative follow-up. None of these comparisons supplies the independence or missingness conditions absent from the clinical observation scheme.

Figure S1 displays the observation schedule and selected trajectories.

For the separate intercept-only analysis in Figure S2, positive observed bilirubin and the 3,000-day restriction leave 1,790 visits from 312 patients. The equal-subject local linear estimate uses $h = 0.09446$, 9,999 multipliers,

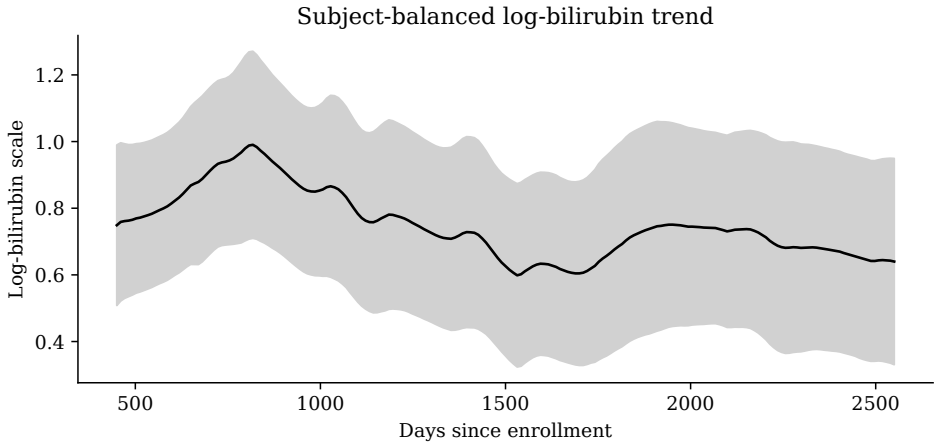


Figure S2: Subject-balanced log-bilirubin trend and nominal 95% band under the working sampling model.

and the same 201-point evaluation grid as the regression. The critical value is 2.787 and average band width is 0.555. The fit describes the weighted observed visits under the working model. Table S3 summarizes the regression bandwidth sensitivity results.

Table S3: PBC nominal joint-band average widths for three bandwidth factors. The primary factor is $c = 1$.

Coefficient	$c = 0.8$	$c = 1$	$c = 1.2$
Intercept	0.774	0.647	0.602
D-penicillamine	1.056	0.899	0.846
Edema	0.593	0.483	0.441
Albumin	0.731	0.621	0.579
Prothrombin time	0.767	0.670	0.629
Log ALP	0.579	0.500	0.467

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